

Chapter 21

Differential Equations

Solutions

SECTION - A

Objective Type Questions (One option is correct)

[Order, Degree and Formation of Differential Equation]

1. If m, n are order and degree of differential equation $y \frac{dy}{dx} + x^3 \left(\frac{d^2y}{dx^2} \right) - xy = \cos x$ then
- (1) $m < n$ (2) $m = n$ (3) $m > n$ (4) $m - n = 3$

Sol. Answer (3)

2. The degree of the differential equation $\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{3/2} = \frac{d^2y}{dx^2}$ is
- (1) 4 (2) $\frac{3}{2}$ (3) Not defined (4) 2

Sol. Answer (4)

3. The degree of the differential equation of the curve $(x - a)^2 + y^2 = 16$ will be
- (1) 0 (2) 2 (3) 3 (4) 1

Sol. Answer (2)

4. If $y = A \sin(\theta + B)$, where A and B are arbitrary constant then to form a differential equation how many times it should be differentiated?
- (1) 1 (2) 2 (3) 3 (4) Cannot be formed

Sol. Answer (2)

5. Which of the following differential equation has $y = x$ as one of its particular solution?

- (1) $\frac{d^2y}{dx^2} - x^2 \frac{dy}{dx} + xy = x$ (2) $\frac{d^2y}{dx^2} - x^2 \frac{dy}{dx} + xy = 0$
- (3) $\frac{d^2y}{dx^2} + x^2 \frac{dy}{dx} + xy = x$ (4) $\frac{d^2y}{dx^2} + x \frac{dy}{dx} + xy = 0$

Sol. Answer (2)

6. The differential equation for which $y = a\cos x + b\sin x$ is a solution is

(1) $\frac{d^2y}{dx^2} + y = 0$ (2) $\frac{d^2y}{dx^2} - y = 0$ (3) $\frac{d^2y}{dx^2} + (a+b)y = 0$ (4) $\frac{d^2y}{dx^2} + (a-b)y = 0$

Sol. Answer (1)

7. Which of the following is general solution of $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = 0$?

(1) $y = (Ax + B)e^x$ (2) $y = (Ax + B)e^{-x}$ (3) $y = A\cos x + B\sin x$ (4) $y = Ae^x + Be^{-x}$

Sol. Answer (1)

8. The order and degree of the differential equation whose general equation is $y = c(x - c)^2$ are

(1) 1, 1 (2) 1, 2 (3) 1, 3 (4) 2, 1

Sol. Answer (3)

9. Family $y = Ax + A^3$ of curves will correspond to a differential equation of order

(1) 3 (2) 2 (3) 1 (4) Not defined

Sol. Answer (3)

10. The degree of the differential equation

$$\left(\frac{d^3y}{dx^3}\right)^{2/3} + 4 - 3\frac{d^2y}{dx^2} + 5\frac{d^3y}{dx^3} = 0 \text{ is}$$

(1) 1 (2) 2 (3) 3 (4) Not defined

Sol. Answer (3)

We have, $\left(\frac{d^3y}{dx^3}\right)^{2/3} + 4 - 3\frac{d^2y}{dx^2} + 5\frac{d^3y}{dx^3} = 0$

$$\Rightarrow \left(\frac{d^3y}{dx^3}\right)^{2/3} = \left(-5\frac{d^3y}{dx^3} + 3\frac{d^2y}{dx^2} - 4\right) \Rightarrow \left(\frac{d^3y}{dx^3}\right)^2 = \left(-5\frac{d^3y}{dx^3} + 3\frac{d^2y}{dx^2} - 4\right)^3$$

Clearly the degree of the differential equation is 3.

11. The differential equation representing the family of curves $y^2 = 2c(x + \sqrt{c})$, where c is a positive parameter, is of

(1) Order 1, degree 3 (2) Order 1, degree 2 (3) Order 2, degree 3 (4) Order 2, degree 2

Sol. Answer (1)

$$y^2 = 2c(x + \sqrt{c}) \quad \dots(i)$$

As only one arbitrary constant is present in the equation hence we can differentiate the equation only once.

Differentiating the given equation w.r.t. 'x' we get

$$2y \frac{dy}{dx} = 2c$$

$$\Rightarrow y \frac{dy}{dx} = c \quad \dots(ii)$$

On solving (i) and (ii) we get

$$y^2 = 2 \cdot \left(y \frac{dy}{dx} \right) \left(x + \sqrt{y \frac{dy}{dx}} \right)$$

$$\Rightarrow y^2 - 2yx \frac{dy}{dx} = 2 \left(y \frac{dy}{dx} \right)^{3/2} \Rightarrow \left(y^2 - 2xy \frac{dy}{dx} \right)^2 = 4 \left(y \frac{dy}{dx} \right)^3$$

Clearly the degree of the differential equation is 3 and order is 1.

12. The differential equation of the family of curves $y = P(x + Q)^2$ is

(1) $yy'' = (y')^2$

(2) $2yy'' = (y')^2$

(3) $2yy'' = y' + y$

(4) $2yy'' = y'' - y$

Sol. Answer (2)

We have,

$$y = P(x + Q)^2 \quad \dots(i)$$

$$\Rightarrow y' = 2P(x + Q) \quad \dots(ii)$$

$$\Rightarrow y'' = 2P \quad \dots(iii)$$

From (ii) and (iii), we get

$$y' = y''(x + Q)$$

$$\Rightarrow x + Q = \frac{y'}{y''} \quad \dots(iv)$$

Now by using (i), (iii) and (iv), we get

$$y = \frac{y''}{2} \left(\frac{y'}{y''} \right)^2$$

$$\Rightarrow 2y''y = (y')^2$$

13. The differential equation of all circles passing through the origin and having their centres on the x-axis is

(1) $y^2 = x^2 - 2xy \frac{dy}{dx}$

(2) $x^2 = y^2 + xy \frac{dy}{dx}$

(3) $x^2 = y^2 + 3xy \frac{dy}{dx}$

(4) $y^2 = x^2 + 2xy \frac{dy}{dx}$

Sol. Answer (4)

Let the radius of the circle is 'a' then its equation is

$$(x - a)^2 + y^2 = a^2 \quad \dots(i)$$

Differentiating it w.r.t. x,

we get

$$2(x - a) + 2y \frac{dy}{dx} = 0 \Rightarrow (x - a) = -y \frac{dy}{dx}$$

$$\Rightarrow a = x + y \frac{dy}{dx} \quad \dots(ii)$$

Equation (i) can be written as

$$x^2 + y^2 - 2ax = 0 \quad \dots(iii)$$

by (ii) and (iii)

$$x^2 + y^2 - 2x \left(x + y \frac{dy}{dx} \right) = 0$$

$$y^2 = x^2 + 2xy \frac{dy}{dx}$$

14. The differential equation of all ellipse center at the origin having major and minor axes along coordinate axes is

$$(1) \quad xyy_2 - xy_1^2 + yy_1 = 0 \quad (2) \quad xyy_2 + xy_1^2 - yy_1 = 0 \quad (3) \quad xyy_2 + xy_1^2 + yy_1 = 0 \quad (4) \quad xyy_2 + xy_1^2 = 0$$

Sol. Answer (2)

Equation of the ellipse can be written as

$$Ax^2 + By^2 = 1 \quad \dots(i)$$

differentiating it w.r.t. x we get

$$2Ax + 2By \frac{dy}{dx} = 0$$

$$Ax + By \frac{dy}{dx} = 0 \quad \dots(ii)$$

again differentiating

$$A + B \left(y \frac{d^2y}{dx^2} + \left(\frac{dy}{dx} \right)^2 \right) = 0 \quad \dots(iii)$$

$$\text{If } \frac{dy}{dx} = y_1 \text{ and } \frac{d^2y}{dx^2} = y_2$$

the equation (i), (ii) and (iii) may be written as

$$Ax^2 + By^2 - 1 = 0 \quad \dots(iv)$$

$$\Rightarrow Ax + Byy_1 = 0 \quad \dots(v)$$

$$\Rightarrow A + B(yy_2 + y_1^2) = 0 \quad \dots(vi)$$

by (v) and (vi)

$$\frac{A}{B} = -\frac{yy_1}{x} = -(yy_2 + y_1^2)$$

$$yy_1 = xyy_2 + xy_1^2$$

$$xyy_2 + xy_1^2 - yy_1 = 0$$

(2) is correct

15. The differential equation of all parabolas with axis parallel to the axis of y is

$$(1) \quad y_2 = 2y_1 + x$$

$$(2) \quad y_3 = 2y_1$$

$$(3) \quad y_2^3 = y_1$$

$$(4) \quad y_3 = 0$$

Sol. Answer (4)

Parabola whose axis is parallel to y -axis is

$$ay^2 + by + c = x \quad \dots(i)$$

where a, b, c are arbitrary constants

$$\text{Differentiating equation (i)} \quad 2ayy_1 + by_1 = 1 \quad \dots(ii)$$

Differentiating again

$$2ayy_2 + 2ay_1^2 + by_2 = 0$$

$$\Rightarrow 2yy_2 + 2y_1^2 + \frac{b}{a}y_2 = 0 \quad \dots(iii)$$

again differentiating

$$2yy_3 + 2y_2y_1 + 4y_1y_2 + \frac{b}{a}y_3 = 0 \quad \dots(iv)$$

by (iii) and (iv)

$$\frac{2yy_2 + 2y_1^2}{2yy_3 + 6y_2y_1} = \frac{y_2}{y_3}$$

$$\Rightarrow \frac{yy_2 + y_1^2}{yy_3 + 3y_2y_1} = \frac{y_2}{y_3}$$

$$\Rightarrow (yy_2 + y_1^2)y_3 - (yy_2y_3 + 3y_2^2y_1) = 0$$

$$\Rightarrow yy_3 + 3y_2^2 = 0$$

16. The second order differential equation is

$$(1) \quad (y')^2 = y^2 - x$$

$$(2) \quad y'y'' + y = \sin x$$

$$(3) \quad y''' + y'' + y = 0$$

$$(4) \quad y' = y$$

Sol. Answer (2)

Clearly the second order differential equation is

$$y'y'' + y = \sin x$$

17. The order of the differential equation whose general solution is $y = (C_1 + C_2)\cos(x + C_3) - C_4e^{x+C_5}$ where C_1, C_2, C_3, C_4 and C_5 are arbitrary constant is

$$(1) \quad 2$$

$$(2) \quad 3$$

$$(3) \quad 4$$

$$(4) \quad 5$$

Sol. Answer (2)

$$\text{We have, } y = (C_1 + C_2) \cos(x + C_3) - C_4 \cdot e^x \cdot e^{C_5}$$

$$y = C \cos(x + C_3) - C'e^x \text{ where } C = C_1 + C_2 \text{ and } C' = C_4 \cdot e^{C_5}$$

Since the equation has three arbitrary constants C, C', C_3 , hence the order of the equation is 3.

(2) is correct.

18. The order of differential equation of family of all concentric circles centered at (h, k) is

- (1) 1 (2) 2 (3) 3 (4) 4

Sol. Answer (1)

$$(x-h)^2 + (y-k)^2 = r^2, \text{ only one arbitrary constant}$$

$$\Rightarrow (x-h) + (y-k) \frac{dy}{dx} = 0 \Rightarrow \text{order} = 1$$

[Solution of Differential Equation]

19. The solution of $\frac{dy}{dx} + y = e^{-x}$, $y(0) = 0$ is

- (1) $y = e^{-x}(x-1)$ (2) $y = xe^x$ (3) $y = xe^{-x} + 1$ (4) $y = xe^{-x}$

Sol. Answer (4)

20. Solution of the differential equation $xdy - ydx = 0$ represents

- (1) A rectangular hyperbola (2) Parabola whose vertex is at origin
(3) Straight line passing through origin (4) A circle whose centre is at origin

Sol. Answer (3)

21. Integrating factor of the differential equation $\cos x \frac{dy}{dx} + y \sin x = 1$ is

- (1) $\cos x$ (2) $\tan x$ (3) $\sec x$ (4) $\sin x$

Sol. Answer (3)

22. Solution of the differential equation, $\tan x \cdot \sec^2 x dx + \tan x \cdot \sec^2 y dy = 0$ is

- (1) $\tan x + \tan y = k$ (2) $\tan x - \tan y = k$ (3) $\frac{\tan x}{\tan y} = k$ (4) $\tan x \cdot \tan y = k$

Sol. Answer (4)

23. Integrating factor of $x \frac{dy}{dx} - y = x^4 - 3x$ is

- (1) x (2) $\log x$ (3) $\frac{1}{x}$ (4) $-x$

Sol. Answer (3)

24. The general solution of differential equation $(e^x + 1)ydy = (y + 1)e^x dx$ is

- (1) $y + 1 = k(e^x + 1)$ (2) $y + 1 = e^x + k$
(3) $y = \log\{k(y + 1)(e^x + 1)\}$ (4) $y = \log\left(\frac{e^x + 1}{y + 1}\right) + k$

Sol. Answer (3)

25. The solution of differential equation $\frac{dy}{dx} = e^{x-y} + x^2 e^{-y}$ is

- (1) $y = e^{x-y} - x^2 e^{-y} + c$ (2) $e^y - e^x = \frac{x^3}{3} + c$ (3) $e^x + e^y = \frac{x^3}{3} + c$ (4) $e^x - e^y = \frac{x^3}{3} + c$

Sol. Answer (2)

26. The solution of differential equation $\frac{dy}{dx} + \frac{2xy}{1+x^2} = \frac{1}{(1+x^2)^2}$ is

(1) $y(1+x^2) = c + \tan^{-1}x$

(2) $\frac{y}{1+x^2} = c + \tan^{-1}x$

(3) $y \log(1+x^2) = c + \tan^{-1}x$

(4) $y(1+x^2) = c + \cos^{-1}x$

Sol. Answer (1)

27. The number of solution of $\frac{dy}{dx} = \frac{y+1}{x-1}$ when $y(1) = 2$ is

(1) None

(2) One

(3) Two

(4) Infinite

Sol. Answer (1)

28. The differential equation $y \frac{dy}{dx} + x = c$ represents

(1) Family of hyperbolas

(2) Family of parabolas

(3) Family of ellipse

(4) Family of circles

Sol. Answer (4)

29. The integrating factor of differential equation $\frac{dy}{dx} + y = \frac{1+y}{x}$ is

(1) $\frac{x}{e^x}$

(2) $\frac{e^x}{x}$

(3) xe^x

(4) e^x

Sol. Answer (2)

30. The differential equation of family of curves $x^2 + y^2 - 2ay = 0$, where a is an arbitrary constant is

(1) $(x^2 - y^2) \frac{dy}{dx} = 2xy$

(2) $2(x^2 + y^2) \frac{dy}{dx} = xy$

(3) $(x^2 - y^2) \frac{dy}{dx} = xy$

(4) $(x^2 + y^2) \frac{dy}{dx} = 2xy$

Sol. Answer (1)

31. The general solution of $\frac{dy}{dx} = 2xe^{x^2-y}$ is

(1) $e^{x^2-y} = c$

(2) $e^{-y} + e^{x^2} = c$

(3) $e^y = e^{x^2} + c$

(4) $e^{x^2+y} = c$

Sol. Answer (3)

32. The solution of the equation $(2y-1)dx - (2x+3)dy = 0$ is

(1) $\left(\frac{2x-1}{2y+3}\right) = k$

(2) $\frac{2y+1}{2x-3} = k$

(3) $\frac{2x+3}{2y-1} = k$

(4) $\frac{2x-1}{2y-1} = k$

Sol. Answer (3)

33. Let $f(x) = \sec x \cdot f'(x)$, $f(0) = 1$, then $f\left(\frac{\pi}{6}\right)$ equal to

(1) $\frac{1}{\sqrt{e}}$

(2) $\sqrt{e}$

(3) $e^{\frac{3}{2}}$

(4) $\frac{1}{2\sqrt{e}}$

Sol. Answer (2)

34. The integrating factor of $(1 + y^2)dx = (\tan^{-1}y - x)dy$ is

- (1) $\tan^{-1}y$ (2) $\tan y$ (3) $e^{\tan^{-1}y}$ (4) $e^{\tan y}$

Sol. Answer (3)

35. The solution of differential equation $\frac{dy}{dx} = \cos(x - y)$

- (1) $y + \cot\left(\frac{x-y}{2}\right) = c$ (2) $x + \cot\left(\frac{x-y}{2}\right) = c$ (3) $x + \tan\left(\frac{x-y}{2}\right) = c$ (4) $x + \tan\left(\frac{x+y}{2}\right) = c$

Sol. Answer (2)

36. $\frac{dy}{dx} = \frac{xy + y}{xy + x}$, then the solution of differential equation is

- (1) $y = xe^x + c$ (2) $y = e^x + c$ (3) $y = cxe^{x-y}$ (4) $y = x + c$

Sol. Answer (3)

37. The differential equation $ydy + xdx = dx$ represents

- (1) A set of circles with centre on x axis (2) A set of concentric circles
(3) A set of ellipse (4) A set of circles with centre on y axis

Sol. Answer (1)

38. The integrating factor of $\cos^2 x \frac{dy}{dx} + y = \tan x$ is

- (1) $e^{\sin x}$ (2) $e^{\cos x}$ (3) $e^{\tan x}$ (4) $e^{\cot x}$

Sol. Answer (3)

39. The integrating factor of $\frac{dy}{dx} + 2y = xe^{4x}$ is

- (1) e^{2x} (2) x^2 (3) e^{4x} (4) e^x

Sol. Answer (1)

40. The general solution of differential equation $\frac{dy}{dx} = e^{\frac{x^2}{2}} + xy$ is

- (1) $y = ce^{\frac{x^2}{2}}$ (2) $y = ce^{\frac{x^2}{2}}$ (3) $y = (x + c)e^{\frac{x^2}{2}}$ (4) $y = (c - x)e^{\frac{x^2}{2}}$

Sol. Answer (3)

41. The solution of differential equation $\cos x \cdot \sin y dx + \sin x \cdot \cos y dy = 0$ is

- (1) $\frac{\sin x}{\sin y} = c$ (2) $\sin x \cdot \sin y = c$ (3) $\sin x + \sin y = c$ (4) $\cos x \cdot \cos y = c$

Sol. Answer (2)

42. The solution of differential equation $\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$ is

- (1) $y = \tan^{-1}x + c$ (2) $\tan^{-1}y = x + c$ (3) $(y-x) = c(1+xy)$ (4) $\tan xy = c$

Sol. Answer (3)

43. The solution of differential equation $ydx + (x + xy)dy = 0$ is

- (1) $y + \log\left(\frac{y}{x}\right) + c = 0$ (2) $y + \log xy + c = 0$ (3) $y - \log xy + c = 0$ (4) $x + \log\left(\frac{x}{y}\right) + c = 0$

Sol. Answer (2)

44. The solution of $\frac{dy}{dx} = \frac{ax+h}{by+k}$ represents a parabola when

- (1) $a = 0, b = 0$ (2) $a = 1, b = 2$ (3) $a = 0, b \neq 0$ (4) $a = 2, b = 1$

Sol. Answer (3)

We have, $\frac{dy}{dx} = \frac{ax+h}{by+k}$

$$\Rightarrow (by+k)dy = (ax+h)dx \quad \Rightarrow \frac{by^2}{2} + ky = \frac{ax^2}{2} + hx + c$$

$$\Rightarrow ax^2 - by^2 + 2hx - 2ky + 2c = 0$$

Clearly for parabola either $a = 0, b \neq 0$ or $a \neq 0, b = 0$.

45. The solution of the equation $2xy' - y = 3$ represents a family of

- (1) Circle (2) Straight line (3) Ellipse (4) Parabola

Sol. Answer (4)

The given differential equation is

$$2xy' - y = 3$$

$$\Rightarrow 2x \frac{dy}{dx} - y = 3$$

$$\Rightarrow \frac{dy}{dx} - \frac{y}{2x} = \frac{3}{2x}, \text{ which is linear in } y$$

$$\therefore \text{I.F.} = e^{\int -\frac{1}{2x} dx} = e^{-\frac{1}{2} \log|x|} = e^{\log_e|x|^{-\frac{1}{2}}} = |x|^{-\frac{1}{2}} = \frac{1}{\sqrt{x}}$$

The solution of the equation is given by

$$y \cdot |x|^{-\frac{1}{2}} = \int \frac{3}{2x} \cdot |x|^{-\frac{1}{2}} dx$$

$$y \cdot x^{-\frac{1}{2}} = \frac{3}{2} \int x^{-\frac{3}{2}} dx + C$$

$$y \cdot x^{-\frac{1}{2}} = \frac{3}{2} \cdot \frac{x^{-\frac{1}{2}}}{-\frac{1}{2}} + C$$

$$y \cdot x^{-\frac{1}{2}} = -3x^{-\frac{1}{2}} + C \Rightarrow y = -3 + Cx^{\frac{1}{2}}$$

$$\Rightarrow (y+3)^2 = C^2 x$$

This is the equation of parabola. (4) is correct.

46. If $\frac{dp}{dy} = 3^{\cos y} \sin y$, then p is equal to

(1) $\sin y + C$

(2) $3^{\cos y} + C$

(3) $\frac{-3^{\cos y}}{\ln 3} + C$

(4) $3^{\sin y} + C$

Sol. Answer (3)

$$\frac{dp}{dy} = 3^{\cos y} \cdot \sin y$$

$$\Rightarrow dp = 3^{\cos y} \sin y \, dy$$

Integrating both sides

$$\int dp = \int 3^{\cos y} \cdot \sin y \, dy$$

$$\Rightarrow p = \int 3^{\cos y} \cdot \sin y \, dy$$

$$\Rightarrow p = -\frac{3^{\cos y}}{\ln 3} + C$$

Option (3) is correct.

47. The solution of $y' - y = 1$, $y(0) = 1$, is given by $y(x) =$

(1) $-\exp(x)$

(2) $-\exp(-x)$

(3) 1

(4) $2\exp(x) - 1$

Sol. Answer (4)

$$\frac{dy}{dx} - y = 1$$

$$\Rightarrow \frac{dy}{1+y} = dx$$

Integrating

$$\ln(1+y) = x + C$$

$$\Rightarrow 1+y = e^{x+C}$$

Initially $x = 0$, $y = 1$

$$\Rightarrow 1+1 = e^C \Rightarrow C = \ln 2$$

Hence we have

$$1 + y = e^{x+\ln 2} = e^x \cdot e^{\ln 2} = 2e^x$$

$$y = 2e^x - 1 = 2\exp(x) - 1$$

Hence option (4) is correct.

48. The solution of the equation $y \frac{dy}{dx} = x - 1$, $y(1) = 1$, is

(1) $y^2 = x^2 + 2x + 2$

(2) $y^2 = 2x^2 + x + 1$

(3) $y^2 = 2x^2 - x - 1$

(4) $y^2 = x^2 - 2x + 2$

Sol. Answer (4)

The given differential equation is

$$y \frac{dy}{dx} = x - 1$$

$$\Rightarrow \int y \, dy = \int (x - 1) \, dx$$

$$\Rightarrow \frac{y^2}{2} = \frac{x^2}{2} - x + c$$

$$\Rightarrow y^2 = x^2 - 2x + 2c$$

Initially at $x = 1$, $y = 1$

putting $x = 1$, $y = 1$ in (i)

$$1 = 1 - 2 + 2c$$

$$\Rightarrow c = 1$$

$$y^2 = x^2 - 2x + 2$$

(4) is correct

49. Solution of differential equation $\frac{dy}{dx} = \frac{2}{x+y}$ is

(1) $x + y + 2 = ke^{y/2}$

(2) $x - y + 2 = ke^{y/2}$

(3) $x + y + 2 = ke^{-y/2}$

(4) $x - y + 2 = ke^{-y/2}$

Sol. Answer (1)

We have, $\frac{dy}{dx} = \frac{2}{x+y}$

$$\Rightarrow \frac{dx}{dy} = \frac{x+y}{2}$$

$$\frac{dx}{dy} - \frac{1}{2}x = \frac{1}{2}y$$

Which is the form $\frac{dx}{dy} + Px = Q$.

$$\text{Integrating factor} = e^{\int -\frac{1}{2} dy} = e^{-\frac{y}{2}}$$

The solution is given by $xe^{\frac{y}{2}} = \int \frac{y}{2} e^{\frac{y}{2}} dy = \frac{y}{2} \frac{e^{\frac{y}{2}}}{\frac{1}{2}} - \int \frac{1}{2} \cdot \frac{e^{\frac{y}{2}}}{\frac{1}{2}} dy = ye^{\frac{y}{2}} - \frac{e^{\frac{y}{2}}}{\frac{1}{2}} + k$

$$\Rightarrow x = -y - 2 + ke^{\frac{y}{2}}$$

$$\Rightarrow x + y + 2 = ke^{\frac{y}{2}}$$

50. If $y = y(x)$ and $\left(\frac{2 + \sin x}{y + 1}\right) \frac{dy}{dx} = -\cos x$, $y(0) = 1$, then $y\left(\frac{\pi}{2}\right) =$

(1) $\frac{1}{3}$

(2) $\frac{2}{3}$

(3) $-\frac{1}{3}$

(4) 1

Sol. Answer (1)

We have, $\left(\frac{2 + \sin x}{y + 1}\right) \frac{dy}{dx} = -\cos x$, $y(0) = 1$

$$\Rightarrow \int \frac{dy}{y+1} = \int \frac{-\cos x}{2 + \sin x} dx$$

$$\Rightarrow \log |y+1| = -\log |2 + \sin x| + c \quad \dots (i)$$

at $x = 0$, $y = 1$

$$\Rightarrow \log |2| = -\log |2 + 0| + c$$

$$\Rightarrow c = 2\log 2$$

Putting the value of (c) the equation (i) becomes

$$\log |y+1| = -\log |\sin x + 2| + 2\log 2$$

$$\Rightarrow (y+1) = \frac{4}{\sin x + 2}$$

at $x = \frac{\pi}{2}$

$$y+1 = \frac{4}{\sin \frac{\pi}{2} + 2} = \frac{4}{3}$$

$$\Rightarrow y = \frac{4}{3} - 1 = \frac{1}{3}$$

$$\Rightarrow y\left(\frac{\pi}{2}\right) = \frac{1}{3}$$

51. The solution of differential equation $(1+x^2)\frac{dy}{dx} + y = e^{\tan^{-1}x}$

$$(1) \quad ye^{\tan^{-1}x} = \frac{1}{2}e^{2\tan^{-1}x} + C$$

$$(2) \quad y = \frac{1}{2}e^{2\tan^{-1}x} + C$$

$$(3) \quad ye^{\tan^{-1}x} = 2e^{2\tan^{-1}x} + C$$

$$(4) \quad y \cdot \tan^{-1}x = \frac{1}{2}e^{2\tan^{-1}x} + C$$

Sol. Answer (1)

$$(1+x^2)\frac{dy}{dx} + y = e^{\tan^{-1}x}$$

$$\frac{dy}{dx} + \frac{y}{1+x^2} = \frac{e^{\tan^{-1}x}}{1+x^2}, \text{ which is linear in } y.$$

$$\therefore \text{ Integrating factor I.F.} = e^{\int \frac{1}{1+x^2} dx} = e^{\tan^{-1}x}$$

The solution of the differential equation is

$$y(e^{\tan^{-1}x}) = \int \frac{e^{\tan^{-1}x}}{1+x^2} \cdot e^{\tan^{-1}x} dx$$

$$y(e^{\tan^{-1}x}) = \frac{(e^{\tan^{-1}x})^2}{2} + C$$

$$ye^{\tan^{-1}x} = \frac{1}{2}e^{2\tan^{-1}x} + C$$

52. The solution of differential equation $x^2y^2dy = (1-xy^3)dx$ is

$$(1) \quad x^3y^3 = x^2 + C$$

$$(2) \quad 2x^3y^3 = 3x^2 + C$$

$$(3) \quad x^3y^3 = x^2 + x + C$$

$$(4) \quad x^3y^3 = 3x^2 + C$$

Sol. Answer (2)

$$x^2y^2dy = (1-xy^3)dx$$

$$\Rightarrow \frac{dy}{dx} = \frac{1-xy^3}{x^2y^2} \Rightarrow \frac{dy}{dx} = \frac{1}{x^2y^2} - \frac{y}{x}$$

$$\Rightarrow \frac{dy}{dx} + \frac{1}{x} \cdot y = \frac{1}{x^2y^2} \Rightarrow x \frac{dy}{dx} + y = \frac{x}{(xy)^2}$$

$$\Rightarrow \left(x \frac{dy}{dx} + y \right) (xy)^2 = x \Rightarrow (xy)^2 d(xy) = x dx$$

$$\Rightarrow \frac{(xy)^3}{3} = \frac{x^2}{2} + k \Rightarrow 2x^3y^3 = 3x^2 + C$$

53. The solution of the differential equation $ydx + (x + x^2y)dy = 0$ is

(1) $\log y = Cx$ (2) $-\frac{1}{xy} + \log y = C$ (3) $\frac{1}{xy} + \log y = C$ (4) $-\frac{1}{xy} = C$

Sol. Answer (2)

The given differential equation is,

$$ydx + (x + x^2y)dy = 0$$

$$\Rightarrow \frac{dx}{dy} + \frac{x}{y} + x^2 = 0$$

$$\Rightarrow \frac{1}{x^2} \frac{dx}{dy} + \left(\frac{1}{x}\right)\left(\frac{1}{y}\right) = -1 \quad \dots(i)$$

$$\text{Let } \frac{1}{x} = v \Rightarrow -\frac{1}{x^2} \frac{dx}{dy} = \frac{dv}{dy}$$

Hence equation (i) becomes

$$-\frac{dv}{dy} + v \cdot \frac{1}{y} = -1$$

$$\Rightarrow \frac{dv}{dy} - \frac{v}{y} = 1$$

$$\Rightarrow \frac{dv}{dy} + v\left(-\frac{1}{y}\right) = 1, \text{ which is linear in } v$$

$$\text{Integrating factor I.F.} = e^{\int -\frac{1}{y} dy} = \frac{1}{y}$$

The solution of the equation is

$$v\left(\frac{1}{y}\right) = \int \frac{1}{y} dy$$

$$\frac{v}{y} = \ln |y| + k$$

$$\Rightarrow -\frac{1}{xy} + \ln |y| = C$$

54. Solution of $y dx - x dy = x^2 y dx$ is

(1) $y^2 e^{x^2} = cx^2$ (2) $ye^{-x^2} = cx^2$ (3) $y''' + y'' + y = 0$ (4) $y' = y$

Sol. Answer (1)

$$y dx - x dy = x^2 y dx$$

$$\Rightarrow x dy + x^2 y dx = y dx$$

$$\Rightarrow x \frac{dy}{dx} + x^2 y = y$$

$$\Rightarrow \frac{dy}{dx} + xy = \frac{y}{x}$$

$$\Rightarrow \frac{dy}{y} + x dx = \frac{1}{x} dx$$

integrating

$$2 \ln y + x^2 = 2 \ln x + \ln c \quad \text{if } x, y > 0$$

$$\Rightarrow y^2 e^{x^2} = cx^2$$

(1) is correct

[Miscellaneous]

55. The real value of n for which substitution $y = u^n$ will transform differential equation $2x^4 y \frac{dy}{dx} + y^4 = 4x^6$, into a homogeneous equation

(1) $\frac{1}{2}$

(2) 1

(3) $\frac{3}{2}$

(4) 2

Sol. Answer (3)

$$y = u^n \Rightarrow \frac{dy}{dx} = nu^{n-1} \frac{du}{dx}$$

$$\Rightarrow 2x^4 \cdot u^n \cdot nu^{n-1} \frac{du}{dx} + u^{4n} = 4x^6$$

$$\Rightarrow \frac{du}{dx} = \frac{4x^6 - u^{4n}}{2nx^4 u^{2n-1}}$$

$\therefore$ It is homogeneous, degree of $4x^6 - u^{4n}$ and $2nx^4 u^{2n-1}$ will be same.

$$\therefore 4n = 6 \text{ and } 4 + 2n - 1 = 6 \Rightarrow n = \frac{3}{2}$$

56. Solution of differential equation $\frac{dy}{dx} \tan y = \sin(x+y) + \sin(x-y)$, is

(1) $\sec y + 2 \cos x = c$

(2) $\sec y - 2 \cos x = c$

(3) $\cos y - 2 \sin x = c$

(4) $\tan y - 2 \sec x = c$

Sol. Answer (1)

$$\frac{dy}{dx} (\tan y) = \sin(x+y) + \sin(x-y)$$

$$\Rightarrow \frac{dy}{dx}(\tan y) = 2 \sin x \cos y$$

$$\therefore \int \tan y \sec y dy = \int 2 \sin x dx$$

$$\therefore \sec y + 2 \cos x = c$$

57. For solving $\frac{dy}{dx} = 4x + y + 1$, suitable substitution is

(1) $y = vx$

(2) $y = 4x$

(3) $y = 4x + v$

(4) $y + 4x + 1 = v$

Sol. Answer (4)

$y + 4x + 1 = v$ is suitable as,

$$\frac{dy}{dx} = f(ax + by + c) \text{ is solvable for substituting } ax + by + c = v$$

58. The integration factor of equation $(x^2 + 1)\frac{dy}{dx} + 2xy = x^2 - 1$, is

(1) $x^2 + 1$

(2) $\frac{2x}{x^2 + 1}$

(3) $\frac{x^2 - 1}{x^2 + 1}$

(4) $1 - x^2$

Sol. Answer (1)

$$(1 + x^2)\frac{dy}{dx} + 2xy = x^2 - 1$$

$$\therefore \frac{dy}{dx} + \left\{ \frac{2x}{1+x^2} \right\} y = \left\{ \frac{x^2 - 1}{x^2 + 1} \right\}$$

$$\therefore \text{IF} = e^{\int \frac{2x}{1+x^2} dx} = e^{\log_e(1+x^2)} = 1+x^2$$

59. A particle starts at origin and moves along the x-axis in such a way that its velocity at point $(x, 0)$ is given by

$$\frac{dx}{dt} = \cos^2 \pi x. \text{ Then particle never reaches point, on}$$

(1) $x = \frac{1}{4}$

(2) $x = \frac{3}{4}$

(3) $x = \frac{1}{2}$

(4) $x = 1$

Sol. Answer (3)

$$\frac{dx}{dt} = \cos^2 \pi x,$$

On differentiating w.r.t. 'x'

$$\frac{d^2x}{dt^2} = -2\pi \sin(2\pi x) = \text{negative}$$

The particle never reaches point, it means

$$\frac{d^2x}{dt^2} = 0 \Rightarrow -2\pi \sin 2\pi x = 0$$

$$\therefore \sin 2\pi x = \sin \pi$$

$$\Rightarrow 2\pi x = \pi \Rightarrow x = \frac{1}{2}$$

60. If $y(t)$ is a solution of $(1+t)\frac{dy}{dt} - ty = 1$ and $y(0) = -1$ then $y(1)$ equals

(1) $-\frac{1}{2}$

(2) $e + \frac{1}{2}$

(3) $e - \frac{1}{2}$

(4) $\frac{1}{2}$

Sol. Answer (1)

$$\frac{dy}{dt} - \left\{ \frac{t}{1+t} \right\} y = \left\{ \frac{1}{1+t} \right\} \text{ and } y(0) = -1$$

$$\text{IF} = e^{-\int \frac{t}{1+t} dt} = e^{-t} \{1+t\}$$

$$\therefore y \cdot e^{-t} \{1+t\} = \int \frac{1}{1+t} e^{-t} (1+t) dt + c$$

$$\therefore ye^{-t}(1+t) = -e^{-t} + c$$

$$y(0) = -1 \Rightarrow c = 0$$

$$\therefore y = -\frac{1}{1+t} \Rightarrow y(1) = -\frac{1}{2}$$

61. A differentiable function $y(x)$ in $(0, \pi)$ satisfying $y' = 1 + y^2$, $y(0) = 0 = y(\pi)$, is

(1) $\tan x$

(2) $x(x - \pi)$

(3) $(x - \pi)(1 - e^x)$

(4) Not possible

Sol. Answer (4)

$$\frac{dy}{dx} = 1 + y^2 \Rightarrow \int \frac{dy}{1+y^2} = \int dx$$

$$\therefore \tan^{-1} y = x + c$$

$$\text{at } x = 0, y = 0 \Rightarrow c = 0 \Rightarrow \tan^{-1} y = x$$

$$\therefore y = \tan x = y(x)$$

But $\tan x$ is not continuous in $(0, \pi)$

So $y(x)$ is not possible in $(0, \pi)$

62. Solve $\left(1 + e^{\frac{x}{y}}\right) dx + e^{\frac{x}{y}} \left(1 - \frac{x}{y}\right) dy = 0$

(1) $k = x + ye^{x/y}$

(2) $k = x - ye^{x/y}$

(3) $k = x^2 + ye^{x/y}$

(4) $k = x^3 + ye^{x^2/y}$

Sol. Answer (1)

Given differential equation is homogeneous in x and y .

Let $x = vy \Rightarrow \frac{dx}{dy} = v + y \frac{dv}{dy}$

$\therefore$ Given equation reduced to

$$(1 + e^v)(v dy + y dv) + e^v(1 - v) dy = 0$$

$$\Rightarrow (1 + e^v)y dv = -[e^v(1 - v) + v(1 + e^v)] dy = -[e^v - ve^v + v + ve^v] dy$$

$$\Rightarrow \frac{-dy}{y} = \frac{1 + e^v}{v + e^v} dv = \frac{d(v + e^v)}{v + e^v}$$

Integrating, we get

$\ln k - \ln y = \ln |v + e^v|$, k being the constant of integration.

$$\Rightarrow k = y(v + e^v) = y \left(\frac{x}{y} + e^{\frac{x}{y}} \right)$$

$$\Rightarrow k = x + ye^{\frac{x}{y}} \text{ is the general solution.}$$

63. If $\phi(x) = \int_a^x (\phi(x))^{-2} dx$ and $\phi(1) = 0$, then $\phi(2)$ is

(1) $3^{1/3}$

(2) $2^{1/2}$

(3) 1

(4) Zero

Sol. Answer (1)

$$\phi(x) = \int_a^x (\phi(x))^{-2} dx$$

$$\Rightarrow \phi'(x) = \frac{1}{(\phi(x))^2}$$

$$\Rightarrow \phi(x) = [3(x - 1)]^{1/3}$$

$$\Rightarrow \phi(2) = 3^{1/3}$$

64. The differential equation for which $\sin^{-1} x + \sin^{-1} y = 1$, is

(1) $\sqrt{1-x^2} dx + \sqrt{1-y^2} dy = 0$

(2) $\sqrt{1-x^2} dy + \sqrt{1-y^2} dx = 0$

(3) $\sqrt{1-x^2} dy - \sqrt{1-y^2} dx = 0$

(4) $\sqrt{1-x^2} dx = \sqrt{1-y^2} dy = 0$

Sol. Answer (2)

$$\sin^{-1} x + \sin^{-1} y = c$$

$$\Rightarrow \frac{1}{\sqrt{1-x^2}} + \frac{1}{\sqrt{1-y^2}} \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{\sqrt{1-x^2}} = -\frac{dx}{\sqrt{1-y^2}}$$

$$\Rightarrow \sqrt{1-x^2} dy + \sqrt{1-y^2} dx = 0$$

65. The solution of $\left(\frac{dy}{dx}\right)^2 + (2x+y)\frac{dy}{dx} + 2xy = 0$, is

(1) $(y+x^2-c_1)(x+\log y+y^2+c_2)=0$

(2) $(y+x^2-c_1)(x-\log y-c_2)=0$

(3) $(y+x^2-c_1)(x+\log y-c_2)=0$

(4) $(y+x^2-c_1)(3x+\log y-c_2)=0$

Sol. Answer (3)

$$\text{We can write, } p^2 + (2x+y)p + 2xy = 0$$

$$\text{where } p = \frac{dy}{dx}$$

$$\Rightarrow (p+2x)(p+y) = 0$$

$$\Rightarrow p+2x = 0$$

or

$$p+y = 0$$

$$\therefore \frac{dy}{dx} = -2x$$

or

$$\frac{dy}{dx} = -y$$

$$\int dy = \int -2x dx$$

or

$$\int \frac{dy}{y} = \int -dx$$

$$\Rightarrow y = -x^2 + c_1$$

or

$$\ln y = -x + c_2$$

$$\therefore y+x^2-c_1=0$$

or

$$\log y + x - c_2 = 0$$

$$\therefore (y+x^2-c_1)(x+\log y-c_2)=0 \text{ is solution}$$

66. The solution of $x \int_0^x y(t) dt = (x+1) \int_0^x t y(t) dt, x > 0$, is

(1) $y = \frac{c}{x^3} e^{-\frac{1}{x}}$

(2) $y = \frac{c}{x^3} e^{\frac{1}{x}}$

(3) $cx^3 e^{-\frac{1}{x}}$

(4) $cx^3 e^{\frac{1}{x}}$

Sol. Answer (1)

Differentiating given equation w.r.t. x ,

$$xy(x) + 1 \int_0^x y(t) dt = (x+1)xy(x) + 1 \int_0^x t y(t) dt$$

$$\text{i.e., } \int_0^x y(t) dt = x^2 y(x) + \int_0^x t y(t) dt$$

Differentiate again w.r.t. x , we get

$$y(x) = x^2 \cdot y'(x) + 2xy(x) + xy(x)$$

$$\text{i.e., } (1-3x)y(x) = \frac{x^2 dy(x)}{dx}$$

$$\therefore \frac{(1-3x)dx}{x^2} = \frac{dy(x)}{y(x)}, \text{ Integrating, we get}$$

$$\Rightarrow y = \frac{c}{x^3} e^{-\frac{1}{x}} \text{ is solution}$$

67. The curve for which the slope of tangent at any point is equal to the ratio of the abscissa to the ordinate of the point is

(1) An ellipse

(2) Parabola

(3) Circle

(4) Rectangular hyperbola

Sol. Answer (4)

68. The slope of tangent at (x, y) to a curve passing through $(2, 1)$ is $\frac{x^2 + y^2}{2xy}$, then the equation of the curve is

(1) $x(x^2 + y^2) = 10$

(2) $x(x^2 - y^2) = 6$

(3) $2(x^2 - y^2) = 6y$

(4) $2(x^2 - y^2) = 3x$

Sol. Answer (4)

We have, $\frac{dy}{dx} = \frac{x^2 + y^2}{2xy}$, which is homogeneous differential equation.

Let $y = vx$

$$\Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx} \Rightarrow v + x \frac{dv}{dx} = \frac{x^2 + v^2 x^2}{2x(vx)}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{1+v^2}{2v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1+v^2}{2v} - v = \frac{1-v^2}{2v}$$

$$\Rightarrow \int \frac{2v dv}{1-v^2} = \int \frac{dx}{x} \quad \Rightarrow \quad -\log |1-v^2| = \ln |x| + c$$

$$\Rightarrow -\log \left| 1 - \left(\frac{y}{x} \right)^2 \right| = \ln |x| + c$$

but at $x = 2, y = 1$

$$-\log \left(1 - \left(\frac{1}{2} \right)^2 \right) = \log |2| + c$$

$$\Rightarrow -\log 3 + \log 4 = \log 2 + c$$

$$\Rightarrow c = -\log 3 + \log 2$$

Let the equation of curve is $-\log \left(1 - \left(\frac{y}{x} \right)^2 \right) = \log |x| + \log 2 - \log 3$

$$\Rightarrow \frac{x^2}{x^2 - y^2} = \frac{2x}{3} \Rightarrow 3x = 2(x^2 - y^2)$$

69. The family whose x and y intercepts of a tangent at any point are respectively double of the x and y co-ordinates of that point is

(1) $x^2 + y^2 = C$

(2) $x^2 - y^2 = C$

(3) $xy = C$

(4) $x^2 y = C$

Sol. Answer (3)

Let the point of contact be (x_1, y_1)

Equation of tangent is

$y - y_1 = m(x - x_1)$, where m is the slope of tangent

$$= m = \left(\frac{dy}{dx} \right)_{(x_1, y_1)} \quad \dots(i)$$

For x -intercept we put $y = 0$, in (i)

$$0 - y_1 = m(x - x_1)$$

$$\Rightarrow x = x_1 - \frac{y_1}{m}$$

but given that

$$x_1 - \frac{y_1}{m} = 2x_1$$

$$\Rightarrow -\frac{y_1}{m} = x_1$$

Now after generalizing the equation, we put $\left(\frac{dy}{dx}\right)_{(x_1, y_1)} = \frac{dy}{dx}$

$$-\frac{y}{\frac{dy}{dx}} = x$$

$$\Rightarrow \frac{dx}{x} = -\frac{dy}{y} \Rightarrow \log |x| = -\log |y| + \log C$$

$$\Rightarrow |xy| = C \Rightarrow xy = \pm C \Rightarrow xy = \text{constant}$$

70. The equation of the curve, slope of whose tangent at any point (h, k) is $2k/h$ and which passes through the point $(1, 1)$ is

(1) $x^2 = y$

(2) $y^2 = x$

(3) $x^2 = 2y$

(4) $y^2 = 2x$

Sol. Answer (1)

Let the curve is $y = f(x)$.

Slope of tangent at (h, k) is $f'(h)$.

But given slope is $\frac{2k}{h}$

$$\Rightarrow f'(h) = \frac{2k}{h}$$

$\Rightarrow$ replacing $f'(h) \rightarrow f'(x)$

$$h \rightarrow x$$

$$k \rightarrow y$$

The equation (i) becomes

$$f'(x) = \frac{2y}{x}$$

$$\text{But } f'(x) = \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \frac{2y}{x}$$

$$\Rightarrow \frac{dy}{y} = 2 \frac{dx}{x}$$

$$\Rightarrow \ln |y| = 2 \ln |x| + c$$

When $x, y > 0 \Rightarrow |x| = x, |y| = y$

hence $\ln y = 2 \ln x + c \quad \dots(ii)$

Since the curve passes through the point $(1, 1)$, hence

$$\ln 1 = 2 \ln 1 + c$$

$$\Rightarrow c = 0$$

Hence equation (ii) becomes

$$\ln y = 2 \ln x$$

$$x^2 = y$$

(1) is correct

71. The equation of curve in which portion of y -axis cut off between origin and tangent varies as cube of abscissa of point of contact is

$$(1) \quad y = \frac{kx^3}{3} + c$$

$$(2) \quad y = \frac{-kx^3}{2} + cx$$

$$(3) \quad y = \frac{-kx^2}{2} + c$$

$$(4) \quad y = \frac{kx^3}{3} + \frac{cx^2}{2}$$

Sol. Answer (2)

Equation of tangent at (x, y) is

$$Y - y = \frac{dy}{dx}(X - x), \text{ for } y\text{-axis, } X = 0$$

$$\Rightarrow Y = y - x \frac{dy}{dx}$$

$$\text{Given } y - x \frac{dy}{dx} \propto x^3 \Rightarrow y - x \frac{dy}{dx} = kx^3$$

$$\therefore \frac{dy}{dx} - \frac{y}{x} = kx^2 \quad \left\{ \text{IF} = \frac{1}{x} \right\}$$

$$\Rightarrow y = \frac{-kx^3}{2} + cx$$

72. A curve $y = f(x)$ passes through point $P(1, 1)$. The normal to curve at point P is $a(y - 1) + (x - 1) = 0$. If slope of tangent at any point on curve is proportional to ordinate at that point, then equation of curve is

$$(1) \quad y = e^{ax} - 1$$

$$(2) \quad y = e^{ax} + 1$$

$$(3) \quad y = e^{ax} + a$$

$$(4) \quad y = e^{a(x-1)}$$

Sol. Answer (4)

Normal at $P(1, 1) \rightarrow ay + x = a + 1$ (given)

$$\therefore \text{Slope of normal at } (1, 1) = -\frac{1}{a}$$

$$\therefore \text{Slope of tangent at } (1, 1) = a$$

Also given, $\frac{dy}{dx} \propto y \Rightarrow \frac{dy}{dx} = ky$

$\therefore \left. \frac{dy}{dx} \right|_{(1,1)} = k = a \Rightarrow \frac{dy}{dx} = ay$

$\therefore \frac{dy}{y} = a dx$

Integrating, $\ln|y| = ax + c$

$\therefore$ It is passing through (1, 1), then $c = -a$

$\Rightarrow n|y| = a(x-1) \Rightarrow |y| = e^{a(x-1)}$

73. The differential equation $\frac{dy}{dx} = \frac{\sqrt{1-y^2}}{y}$ determines a family of circles with

[IIT-JEE 2007]

- (1) Variable radii and a fixed centre at (0, 1) (2) Variable radii and a fixed centre at (0, -1)
 (3) Fixed radius 1 and variable centers along the x-axis (4) Fixed radius 1 and variable centres along the y-axis

Sol. Answer (3)

$$\frac{dy}{dx} = \frac{\sqrt{1-y^2}}{y}$$

$$\Rightarrow \int \frac{y dy}{\sqrt{1-y^2}} = \int dx$$

$$\Rightarrow -\frac{1}{2} \int \frac{d(\sqrt{1-y^2})}{\sqrt{1-y^2}} = \int dx$$

$$\Rightarrow -\frac{1}{2} \times 2\sqrt{1-y^2} = \int x + k$$

$$\Rightarrow \sqrt{1-y^2} = x + k$$

$$\Rightarrow 1 - y^2 = (x + k)^2$$

$$\Rightarrow (x + k)^2 + y^2 = 1$$

74. Let f be a non-negative function defined on the interval $[0, 1]$. If $\int_0^x \sqrt{1-(f'(t))^2} dt = \int_0^x f(t) dt$, $0 \leq x \leq 1$, and $f(0) = 0$, then

[IIT-JEE 2009]

(1) $f\left(\frac{1}{2}\right) < \frac{1}{2}$ and $f\left(\frac{1}{3}\right) > \frac{1}{3}$

(2) $f\left(\frac{1}{2}\right) > \frac{1}{2}$ and $f\left(\frac{1}{3}\right) > \frac{1}{3}$

(3) $f\left(\frac{1}{2}\right) < \frac{1}{2}$ and $f\left(\frac{1}{3}\right) < \frac{1}{3}$

(4) $f\left(\frac{1}{2}\right) > \frac{1}{2}$ and $f\left(\frac{1}{3}\right) < \frac{1}{3}$

Sol. Answer (3)

We have

$$\int_0^x \sqrt{1 - (f'(t))^2} dt = \int_0^x f(t) dt, \quad 0 \leq x \leq 1$$

Differentiating both sides w.r.t. x we get

$$\sqrt{1 - (f'(x))^2} = f(x)$$

$$\Rightarrow 1 - [f'(x)]^2 = [f(x)]^2$$

$$\Rightarrow \left(\frac{dy}{dx}\right)^2 = 1 - y^2 \quad \left[\text{put } y = f(x) \text{ \& } \frac{dy}{dx} = f'(x) \right]$$

$$\Rightarrow \int \frac{1}{\sqrt{1 - y^2}} dy = \pm \int 1 dx$$

$$\Rightarrow \sin^{-1} y = c \pm x$$

$$\Rightarrow y = \sin(c \pm x)$$

$$\Rightarrow f(x) = \sin(c \pm x)$$

$$\Rightarrow f(0) = \sin c$$

$$[f(0) = 0]$$

$$\Rightarrow 0 = \sin c$$

$$\Rightarrow c = 0$$

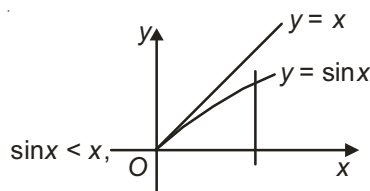
$$\text{Hence } f(x) = \sin(\pm x) = \pm \sin x$$

But $f(x) = -\sin x$ is rejected as f is given to be non-negative

$$\text{Hence } y = \sin x$$

$$\text{For } x \in (0, 1)$$

$$\therefore \sin \frac{1}{2} < \frac{1}{2} \text{ \& } \sin \frac{1}{3} < \frac{1}{3}$$



75. A curve passes through the point $\left(1, \frac{\pi}{6}\right)$. Let the slope of the curve at each point (x, y) be $\frac{y}{x} + \sec\left(\frac{y}{x}\right)$, $x > 0$.

Then the equation of the curve is

[JEE(Advanced)-2013]

$$(1) \sin\left(\frac{y}{x}\right) = \log x + \frac{1}{2}$$

$$(2) \operatorname{cosec}\left(\frac{y}{x}\right) = \log x + 2$$

$$(3) \sec\left(\frac{2y}{x}\right) = \log x + 2$$

$$(4) \cos\left(\frac{2y}{x}\right) = \log x + \frac{1}{2}$$

Sol. Answer (1)

$$\frac{dy}{dx} = \frac{y}{x} + \sec \frac{y}{x}$$

Put $y = vx$

$$\therefore v + x \frac{dv}{dx} = v + \sec v$$

$$\Rightarrow \cos v \, dv = \frac{dx}{x}$$

$$\Rightarrow \sin v = \ln x + c$$

$$\therefore \text{It passes through } \left(1, \frac{\pi}{6}\right)$$

$$\therefore \frac{1}{2} = c$$

$$\therefore \sin \frac{y}{x} = \ln x + \frac{1}{2}$$

76. The function $y = f(x)$ is the solution of the differential equation $\frac{dy}{dx} + \frac{xy}{x^2-1} = \frac{x^4+2x}{\sqrt{1-x^2}}$ in $(-1, 1)$ satisfying

$$f(0) = 0. \text{ Then } \int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x) dx \text{ is}$$

[JEE(Advanced)-2014]

$$(1) \quad \frac{\pi}{3} - \frac{\sqrt{3}}{2}$$

$$(2) \quad \frac{\pi}{3} - \frac{\sqrt{3}}{4}$$

$$(3) \quad \frac{\pi}{6} - \frac{\sqrt{3}}{4}$$

$$(4) \quad \frac{\pi}{6} - \frac{\sqrt{3}}{2}$$

Sol. Answer (2)

$$\frac{dy}{dx} + y \left(\frac{x}{1-x^2} \right) = \frac{x^4+2x}{\sqrt{1-x^2}}$$

$$\text{I.F.} = e^{-\int \frac{x}{1-x^2} dx} = \sqrt{1-x^2}$$

$$\text{So } y\sqrt{1-x^2} = \int (x^4+2x)dx = \frac{x^5}{5} + x^2 + c$$

$$\text{Given } f(0) = 0 \Rightarrow c = 0$$

$$\text{So } y = \frac{\frac{x^5}{5} + x^2}{\sqrt{1-x^2}} = f(x)$$

$$I = \int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x) dx = \int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} \frac{\frac{\sqrt{3}}{2} x^5 + x^2}{\sqrt{1-x^2}} dx = 2 \int_0^{\frac{\sqrt{3}}{2}} \frac{x^2 dx}{\sqrt{1-x^2}}$$

Let $x = \sin \theta$

$$I = 2 \int_0^{\frac{\pi}{3}} \sin^2 \theta d\theta = \int_0^{\frac{\pi}{3}} (1 - \cos 2\theta) d\theta = \left(\theta - \frac{\sin 2\theta}{2} \right) \Big|_0^{\frac{\pi}{3}} = \frac{\pi}{3} - \frac{\sqrt{3}}{4}$$

77. If $y = y(x)$ satisfies the differential equation $8\sqrt{x}(\sqrt{9+\sqrt{x}}) dy = (\sqrt{4+\sqrt{9+\sqrt{x}}})^{-1} dx$, $x > 0$ and $y(0) = \sqrt{7}$, then $y(256) =$ [JEE(Advanced)-2017]

- (1) 80 (2) 9 (3) 16 (4) 3

Sol. Answer (4)

$$\text{As, } dy = \frac{dx}{8\sqrt{x}\sqrt{9+\sqrt{x}} \cdot \sqrt{4+\sqrt{9+\sqrt{x}}}}$$

Integrating,

$$y = \sqrt{4+\sqrt{9+\sqrt{x}}} + c$$

$$\text{at } x = 0, y = \sqrt{7} \Rightarrow c = 0$$

$$\text{so, } y = \sqrt{4+\sqrt{9+\sqrt{x}}}$$

$$\text{at } x = 256, y = 3$$

SECTION - B

Objective Type Questions (More than one options are correct)

1. The foci of the curve which satisfies the equation $(1+y^2)dx - xy dy = 0$ and passes through the point $(1,0)$ are

- (1) $(\sqrt{2}, 0)$ (2) $(0, \sqrt{2})$ (3) $(-\sqrt{2}, 0)$ (4) $(0, -\sqrt{2})$

Sol. Answer (1, 3)

$$(1+y^2) dx - xy dy = 0$$

$$\Rightarrow \frac{dx}{x} = \frac{y dy}{1+y^2}$$

$$\Rightarrow \ln x = \frac{1}{2} \ln(1+y^2) + \ln c, \text{ on integration}$$

$$\Rightarrow 2 \ln x = \ln(1+y^2) + 2 \ln c$$

$$\Rightarrow x^2 = (1+y^2)c^2$$

$$\text{or } 1 + y^2 = \frac{x^2}{c^2}$$

$$\text{at } x = 1, y = 0 \Rightarrow 1 = \frac{1}{c^2} \Rightarrow c^2 = 1$$

Hence the equation becomes

$$1 + y^2 = x^2$$

$$x^2 - y^2 = 1$$

which is a rectangular hyperbola whose eccentricity $e = \sqrt{1^2 + 1^2} = \sqrt{2}$ and transverse axis $2a = 2 \Rightarrow a = 1$

$$\text{foci} \equiv (\pm ae, 0)$$

$$= (\pm \sqrt{2}, 0)$$

Hence option (1), (3) are correct.

2. The general solution of the equation, $x \left(\frac{dy}{dx} \right) = y \cdot \ln(y/x)$ is

$$(1) y = xe^{1-cx}$$

$$(2) y = xe^{1+cx}$$

$$(3) y = xe^{-e^{cx}}$$

$$(4) y = xe^{cx}$$

Sol. Answer (1, 2, 3)

$$x \frac{dy}{dx} = y \ln \left(\frac{y}{x} \right) \Rightarrow \frac{dy}{dx} = \frac{y}{x} \ln \left(\frac{y}{x} \right) \quad \dots(i)$$

Let $y = vx$

Differentiating with respect to x

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

After substitution $y = vx$ and $\frac{dy}{dx} = v + x \frac{dv}{dx}$, the equation (i) becomes

$$v + x \frac{dv}{dx} = v \ln v$$

$$\Rightarrow x \frac{dv}{dx} = v \ln v - v = v (\ln v - 1)$$

$$\int \frac{dv}{v(\ln v - 1)} = \int \frac{dx}{x}$$

Let $(\ln v - 1) = t$

$$\Rightarrow \frac{1}{v} dv = dt$$

After substitution we have

$$\int \frac{dt}{t} = \int \frac{dx}{x}$$

$$\Rightarrow \ln t = \ln x + \ln c \text{ if } t \text{ and } x \text{ both are positive}$$

$$\Rightarrow t = cx \text{ where 'c' is arbitrary constant}$$

$$\Rightarrow (\ln v - 1) = cx$$

$$\Rightarrow \ln v = cx + 1$$

$$\Rightarrow \ln \left(\frac{y}{x} \right) = cx + 1$$

$$\frac{y}{x} = e^{(cx+1)}$$

$$\Rightarrow y = x e^{(cx+1)} = x \cdot e \cdot e^{cx}$$

If we take $-c$ in place of c then $y = x e^{1-cx}$.

Hence option (1), (2), (3) are correct.

3. The equation of the curve passing through (3, 4) and satisfying the differential equation, $y \left(\frac{dy}{dx} \right)^2 + (x - y) \frac{dy}{dx} - x = 0$ can be

(1) $x - y + 1 = 0$

(2) $x^2 + y^2 = 25$

(3) $x^2 + y^2 - 5x - 10 = 0$

(4) $x + y - 7 = 0$

Sol. Answer (1, 2)

The given differential equation is $y \left(\frac{dy}{dx} \right)^2 + (x - y) \frac{dy}{dx} - x = 0$

$$\Rightarrow \frac{dy}{dx} = \frac{-(x - y) \pm \sqrt{(x - y)^2 - 4y(-x)}}{2y}$$

$$\Rightarrow \frac{dy}{dx} = \frac{-(x - y) \pm \sqrt{(x + y)^2}}{2y}$$

$$\Rightarrow \frac{dy}{dx} = \frac{-(x - y) \pm (x + y)}{2y}$$

Case (i) : taking positive sign

$$\frac{dy}{dx} = \frac{-(x - y) + (x + y)}{2y} = \frac{-x + y + x + y}{2y} = 1$$

$$\Rightarrow \frac{dy}{dx} = 1$$

$$\Rightarrow dy = dx$$

Integrating, we get $y = x + c$

but curve passing through (3, 4)

$$4 = 3 + c$$

$$c = 1$$

$$\Rightarrow y = x + 1$$

$$x - y + 1 = 0$$

Case (ii) : taking negative sign

$$\frac{dy}{dx} = \frac{-x + y - x - y}{2y} = -\frac{2x}{2y} = -\frac{x}{y}$$

$$y \, dy + x \, dx = 0$$

Integrating

$$\frac{y^2}{2} + \frac{x^2}{2} = c \quad \dots(i)$$

but curve passes through (3, 4)

hence putting $x = 3$ and $y = 4$ in (i)

$$\frac{16}{2} + \frac{9}{2} = c$$

$$\Rightarrow c = \frac{25}{2}$$

In this way equation of the curve becomes

$$x^2 + y^2 = 25$$

Hence option (1), (2) are correct.

4. The graph of the function $y = f(x)$ passing through the point (0, 1) and satisfying the differential equation

$$\frac{dy}{dx} + y \cos x = \cos x \text{ is such that}$$

- (1) It is a differentiable function $\forall x \in R$
- (2) It is continuous $\forall x \in R$
- (3) It is periodic
- (4) It is passing through $(\pi, 1)$

Sol. Answer (1, 2, 3, 4)

$$\frac{dy}{dx} = -(\cos x)(y - 1)$$

$$\frac{dy}{y-1} = -\cos x \, dx$$

On integration

$$\int \frac{dy}{y-1} = \int -\cos x \, dx$$

$$\Rightarrow \ln |y-1| = -\sin x + c \quad \Rightarrow |y-1| = e^{-\sin x + c}$$

at $x = \frac{\pi}{2}$ and $y = 0$

$$1 = e^{-\sin \frac{\pi}{2} + c}$$

$$\Rightarrow c = 1$$

hence the equation of curve becomes

$$y - 1 = \pm e^{-\sin x + 1}$$

which is periodic, continuous and differentiable and passes through the point $\left(\frac{5\pi}{2}, 0\right)$.

5. Orthogonal trajectories of the system of curves $\left(\frac{dy}{dx}\right)^2 = \frac{a}{x}$ are

$$(1) \quad 9a(y+c)^2 = 4x^3 \quad (2) \quad y+c = \frac{-2}{9\sqrt{a}} x^{3/2} \quad (3) \quad y^2+c = \frac{2}{3\sqrt{a}} x^{3/2} \quad (4) \quad 9a(y+c)^2 = 4x^2$$

Sol. Answer (1)

$$\frac{dy}{dx} = \pm \sqrt{\frac{a}{x}} = \pm \frac{\sqrt{a}}{\sqrt{x}}$$

... (i)

The differential equation of orthogonal trajectory of (i) is obtained by replacing $\frac{dy}{dx} \rightarrow -\frac{dx}{dy}$ hence by (i) $-\frac{dx}{dy} = \frac{\sqrt{a}}{\sqrt{x}}$

$$\Rightarrow \int -\sqrt{x} \, dx = \int \sqrt{a} \, dy$$

$$\Rightarrow -\frac{x^{3/2}}{\frac{3}{2}} = \sqrt{a} \cdot y + c_1 \quad \Rightarrow \quad -\frac{2}{3} x^{3/2} = \sqrt{a} \cdot y + c_1 = \sqrt{a} \left(y + \frac{c_1}{\sqrt{a}} \right)$$

Squaring, we get

$$4x^3 = 9a \left(y + \frac{c_1}{\sqrt{a}} \right)^2$$

Let $\frac{c_1}{\sqrt{a}} = c$ a new arbitrary constant

$$9a(y+c)^2 = 4x^3$$

Hence option (1) is correct.

6. A curve has the property that area of triangle formed by the x-axis, the tangent to the curve and radius vector of the

point of tangency is k^2 . The equation of all such curves passing through (0, 1) is $\ln(ay) = \frac{xy^b}{2k^2}$ then

$$(1) \quad a = 1 \quad (2) \quad b = 1 \quad (3) \quad a = 2 \quad (4) \quad b = 2$$

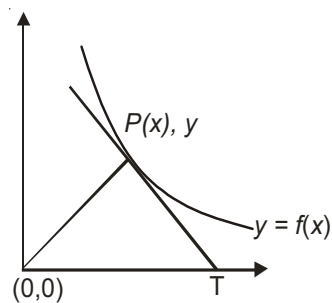
Sol. Answer (1, 2)

Let $P(x, y)$ be a point on the curve, OP is radius vector. PT is tangent to curve at P .

The slope of tangent at P is $\frac{dy}{dx}$

Equation of tangent at $P(x, y)$ is $Y - y = \frac{dy}{dx}(X - x)$

$$Y = 0 \Rightarrow \frac{dy}{dx}(X - x) = -y \Rightarrow X = -\frac{y}{\frac{dy}{dx}} + x$$



$$T \equiv \left(x - y \frac{dx}{dy}, 0 \right)$$

Area of the triangle OPT

$$= \frac{1}{2} \begin{vmatrix} x & y \\ x - y \frac{dx}{dy} & 0 \end{vmatrix}$$

$$\Rightarrow \frac{1}{2} \left[y \left(y \frac{dx}{dy} - x \right) \right] = k^2 \quad (\text{given})$$

$$\Rightarrow y^2 \frac{dx}{dy} - xy = 2k^2$$

$$\frac{dx}{dy} - \left(\frac{1}{y} \right) x = \frac{2k^2}{y^2}$$

$$\text{Integrating factor (I.F.)} = e^{\int P dy} = e^{\int -\frac{1}{y} dy} = \frac{1}{y}$$

So, the solution is $xy = 2k^2 \ln y + c$ But $(0, 1)$ lies on it.

$$\Rightarrow c = 0$$

$$\text{And the equation of the curve is } \ln y = \frac{xy}{2k^2}$$

7. The tangent at any point P of a curve C meets the x -axis at Q whose abscissa is positive and $OP = OQ$, O being the origin, the equation of curve C satisfying these conditions may be

$$(1) \quad y^2 = \frac{1}{4}(1-4x) \quad (2) \quad y^2 = \frac{1}{4}(9-12x) \quad (3) \quad y^2 = \frac{1}{4}(16-16x) \quad (4) \quad y^2 = \frac{1}{4}(25-20x)$$

Sol. Answer (1, 2, 3, 4)

Let the curve be $y = f(x)$. Equation of tangent at any point $P(x, f(x))$ is

$$(Y - f(x)) = f'(x)(X - x)$$

$$\Rightarrow Q = \left(x - \frac{f(x)}{f'(x)}, 0 \right)$$

$$\text{We have } x - \frac{f(x)}{f'(x)} = \sqrt{x^2 + y^2}$$

$$\Rightarrow \frac{y}{\frac{dy}{dx}} = x - \sqrt{x^2 + y^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y \left(x + \sqrt{x^2 + y^2} \right)}{\left(x - \sqrt{x^2 + y^2} \right) \left(x + \sqrt{x^2 + y^2} \right)} = -\frac{x + \sqrt{x^2 + y^2}}{y}$$

$$\Rightarrow \frac{dy}{dx} + \frac{x}{y} = -\frac{\sqrt{x^2 + y^2}}{y}$$

$$\Rightarrow ydy + xdx = -\sqrt{x^2 + y^2} dx$$

$$\Rightarrow \frac{2xdx + 2ydy}{\sqrt{x^2 + y^2}} + 2dx = 0$$

$$\Rightarrow \frac{d(x^2 + y^2)}{\sqrt{x^2 + y^2}} + 2dx = 0$$

Integrating both sides,

$$2\sqrt{x^2 + y^2} + 2x = c$$

$$\Rightarrow 4(x^2 + y^2) = (c - 2x)^2 = 4x^2 + c^2 - 4xc$$

$$\Rightarrow y^2 = \frac{1}{4}(c^2 - 4xc)$$

Now for different value of c we have different parabolas.

8. Consider a curved mirror $y = f(x)$ passing through $(8, 6)$ having the property that all light rays emerging from origin, after getting reflected from the mirror becomes parallel to x -axis. The equation of the mirror is $y^a = b(c - x^d)$ where a, b, c, d are constants, then

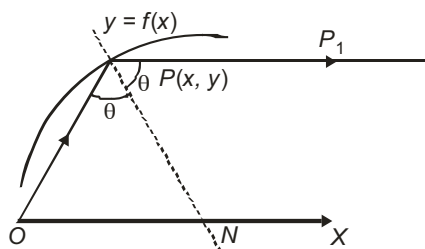
(1) $b = 4$

(2) $b = 36$

(3) $c = 9$

(4) $c = 1$

Sol. Answer (2, 3)



Let the equation of the curved mirror be $y = f(x)$ with the property that any light ray (OP) emerging from origin after getting reflected become parallel to x -axis. If normal drawn at ' P ' meets the x -axis at N , $\angle OPN = \angle NPP_1 = \theta$ (say)

$$\Rightarrow \angle ONP = \theta \Rightarrow OP = ON$$

Equation of normal at P is ;

$$(Y - f(x)) = -\frac{1}{f'(x)}(X - x)$$

$$N \equiv (x + f(x) \cdot f'(x), 0) \text{ or } \left(x + y \frac{dy}{dx}, 0 \right)$$

Since $ON = OP$

$$\sqrt{x^2 + y^2} = \left| x + y \frac{dy}{dx} \right|$$

$$\Rightarrow \frac{2x dx + 2y dy}{\sqrt{x^2 + y^2}} = \pm 2 dx \Rightarrow \frac{d(x^2 + y^2)}{\sqrt{x^2 + y^2}} = \pm 2 dx$$

Integrating both sides we get

$$\Rightarrow 2\sqrt{x^2 + y^2} = \pm 2x + 2k \quad ('k' \text{ is the constant of integration}).$$

$$\Rightarrow \sqrt{x^2 + y^2} = \pm x$$

Curves passes through (8, 6)

$$10 = \pm 8 + k \Rightarrow k = 2, 18$$

$$\text{Thus curve is } \sqrt{x^2 + y^2} = x + 2 \text{ or } \sqrt{x^2 + y^2} = -x + 18$$

$$\text{i.e., } y^2 = 4 + 4x$$

$$\Rightarrow y^2 = 324 - 36x \text{ or } y^2 = 36(9 - x)$$

$$\Rightarrow y^2 = 4(1 + x)$$

9. The equation of all possible curve that might be orthogonal to the family of curves represented by $x^2 + y^2 - 2cx = 0$ (where c is a parameter) may be

$$(1) \quad x^2 + y^2 - ky = 0 \quad (2) \quad x^2 + y^2 + ky = 0 \quad (3) \quad x^2 + y^2 + kx = 0 \quad (4) \quad x^2 + y^2 - kx = 0$$

Sol. Answer (1, 2)

$$\text{Given family of circles is } x^2 + y^2 - 2cx = 0 \quad \dots(1)$$

$$\text{Differentiating it we get; } 2x + 2y \frac{dy}{dx} - 2c = 0$$

Putting the value of $2c$ from (1) in the given family of circles we get;

$$x^2 + y^2 = x \left(2x + 2y \frac{dy}{dx} \right)$$

$$\Rightarrow x^2 + 2xy \frac{dy}{dx} = y^2$$

Now, differentiating equation of required family of curves can be obtained by replacing

$$\frac{dy}{dx} \rightarrow -\frac{dx}{dy}$$

$$\Rightarrow \text{The differential equation of the required family is } x^2 - 2xy \frac{dx}{dy} = y^2$$

$$\Rightarrow x^2 dy - 2xy dx = y^2 dy$$

$$\Rightarrow \frac{y \cdot 2x dx - x^2 dy}{y^2} + dy = 0$$

$$\Rightarrow d\left(\frac{x^2}{y}\right) + dy = 0$$

Integrating we get

$$\frac{x^2}{y} + y = k$$

$$\text{or } x^2 + y^2 - yk = 0, \text{ it can be also written as } x^2 + y^2 + ky = 0$$

10. If a mothball loses volume by evaporation at a rate proportional to its instantaneous area. If the diameter of the ball decreases from 2 cm to 1 cm in 3 months, so the time before which the ball has practically gone is less than or equal to

- (1) 4 months (2) 3 months (3) 2 months (4) 1 month

Sol. Answer (1, 2, 3)

Let at any instant (t), the radius of the mothball be r and its volume be V

$$\Rightarrow V = \frac{4}{3} \pi r^3$$

$$\Rightarrow \frac{dV}{dt} = 4\pi r^2 \cdot \frac{dr}{dt}$$

$$\text{Thus as per information } 4\pi r^2 \cdot \frac{dr}{dt} = -k(4\pi r^2); \text{ where } k \in R^+$$

$$\Rightarrow \frac{dr}{dt} = -k$$

$$\Rightarrow r = -kt + c$$

But at $t = 0$, $r = 2$ cm; $t = 3$ month; $r = 1$ cm.

$$\Rightarrow c = 2; \quad k = \frac{1}{2}$$

$$\Rightarrow r = -\frac{1}{3}t + 2$$

$$\text{Now for } r \rightarrow 0, t \rightarrow \frac{3}{2}$$

Hence it will be one and half month before the ball is practically gone.

11. Let $x(1-x)\frac{dy}{dx} = x-y$

(1) General solution of given differential equation is $xy = (1-x) \ln|1-x| + 1 + c(1-x)$

(2) General solution of given differential equation is $xy = (1-x) \ln|1-x| - 1 + cx(1-x)$

(3) If $y = f(x)$ is a solution of given differential equation, then $\lim_{x \rightarrow 1} f(x)$ does not exist

(4) If $y = f(x)$ is solution of given differential equation then $\lim_{x \rightarrow 1} f(x) = 1$

Sol. Answer (1, 4)

$$\frac{dy}{dx} + \frac{y}{x(1-x)} = \frac{x}{x(1-x)}$$

$$\text{I.F.} = e^{\int \frac{1}{x(1-x)} dx} = e^{\int \left(\frac{1}{x} - \frac{1}{1-x} \right) dx} = \frac{x}{1-x}$$

$$\therefore \text{Solution is } \frac{xy}{1-x} = \int \frac{x}{1-x} \cdot \frac{1}{1-x} dx + c = -\int \frac{1-x-1}{(1-x)^2} dx + c = \ln|1-x| + \frac{1}{1-x} + c$$

$$\Rightarrow xy = (1-x) \ln|1-x| + 1 + c(1-x)$$

$$y = f(x) = \frac{(1-x)}{x} \ln(1-x) + \frac{1}{x} + \frac{c(1-x)}{x}$$

$$\Rightarrow \lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} \left[\frac{1}{x} \cdot \frac{\ln(1-x)}{1/(1-x)} + 1 + 0 \right] = \lim_{x \rightarrow 1} \frac{\frac{1}{x} \times \frac{1}{1-x} \times -1}{-\frac{1}{(1-x)^2} \times -1} + 1 = 0 + 1$$

$$\therefore \lim_{x \rightarrow 1} f(x) = 1$$

12. Let a curve passes through (3, 2) and satisfied the differential equation $(x-1)dx + 4(y-2)dy = 0$

(1) It represents equation of a circle

(2) It represents equation of an ellipse

(3) Area enclosed by the curve is 2π

(4) Line $y = 1$ is a tangent to the curve

Sol. Answer (2, 3, 4)

$$(x-1)dx + 4(y-2)dy = 0$$

$$\frac{(x-1)^2}{2} + \frac{4(y-2)^2}{2} = \text{constant}$$

$$(x-1)^2 + 4(y-2)^2 = k \rightarrow \text{Ellipse } k > 0$$

It passes through (3, 2)

$$\Rightarrow 4 + 0 = k$$

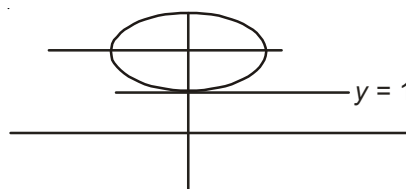
$$\therefore \text{Equation of ellipse is } (x-1)^2 + 4(y-2)^2 = 4$$

$$\text{Area of ellipse} = \pi ab$$

$$= 2\pi$$

Clearly from figure $y = 1$ is a horizontal tangent to the curve.

$\therefore$ Options (2), (3), (4) are correct.



13. A curve satisfies the differential equation $\frac{dy}{dx} = \frac{x+1-xy^2}{x^2y-y}$ and passes through (0, 0)

(1) The equation of the curve is $x^2 + y^2 + 2x = x^2y^2$

(2) The equation of the curve is $x^2 + y^2 + 2x + 2y = x^2y^2$

(3) $x = 0$ is a tangent to curve

(4) $y = 0$ is a tangent to curve

Sol. Answer (1, 3)

$$(x^2y - y)dy = ((x+1) - xy^2)dx$$

$$x^2y dy + xy^2 dx = ydy + xdx + dx$$

$$xy(d(xy)) = xdx + ydy + dx$$

$$\Rightarrow \frac{x^2y^2}{2} = \frac{x^2}{2} + \frac{y^2}{2} + x + c$$

$$\text{It passes through } (0, 0) \Rightarrow c = 0$$

$$\therefore x^2y^2 = x^2 + y^2 + 2x \text{ is the equation of curve clearly } x = 0 \text{ is a tangent at origin.}$$

14. Tangent is drawn at any point P of a curve which passes through (1, 1) cutting x -axis and y -axis at A and B respectively. If $AP : BP = 3 : 1$, then,

(1) Differential equation of the curve is $3x \frac{dy}{dx} + y = 0$

(2) Differential equation of the curve is $3x \frac{dy}{dx} - y = 0$

(3) Curve is passing through $\left(\frac{1}{8}, 2\right)$

(4) Normal at (1, 1) is $x + 3y = 4$

Sol. Answer (1, 3)

Let the tangent be

$$Y - y = \frac{dy}{dx}(X - x)$$

$$A: \left(x - y \frac{dx}{dy}, 0\right) \text{ \& } B: \left(0, y - x \frac{dy}{dx}\right)$$

$$(x, y) = \left(\frac{0 + x - y \frac{dx}{dy}}{4}, \frac{3 \left(y - x \frac{dy}{dx}\right) + 0}{4} \right)$$

$$\Rightarrow 4y = 3y - 3x \frac{dy}{dx} = 3x \frac{dy}{dx} + y = 0$$

Now, solution of differential equation is

$$\Rightarrow 3 \ln y + \ln x = \ln c$$

It passes through (1, 1)

$$\Rightarrow c = 1$$

$$\Rightarrow \text{Equation of curve is } y^3 x = 1$$

$$\text{at } x = \frac{1}{8}, y = 2, \text{ i.e., it passes through } \left(\frac{1}{8}, 2\right)$$

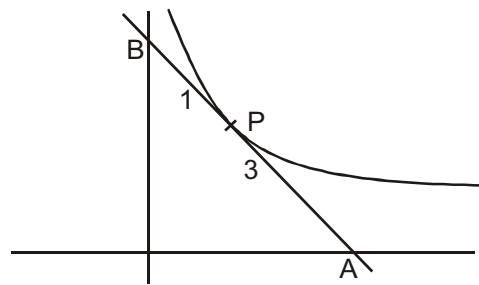
also,

$$\text{Slope of normal at } (1, 1) = \left(+ \frac{3x}{y} \right)_{\text{at } (1, 1)} = 3$$

$\therefore$ Equation of normal is

$$y - 1 = 3(x - 1)$$

$$\Rightarrow 3x - y - 2 = 0$$



15. $y = c_1 x + c_2 \sin(2x + c_3)$ (c_1, c_2, c_3 are arbitrary constants)

(1) Order of differential equation is 2

(2) Order of differential equation is 3

(3) Degree of differential equation is 1

(4) The differential equation is $\frac{y d^3 y}{dx^3} = \frac{dy}{dx} \cdot \frac{d^2 y}{dx^2}$

Sol. Answer (2, 3)

$$y' = c_1 + 2c_2 \cos(2x + c_3)$$

$$y'' = -4c_2 \sin(2x + c_3)$$

$$\Rightarrow y'' = -4c_2 y$$

$$\Rightarrow y''' = -4c_2 y'$$

$$\Rightarrow \frac{y'''}{y''} = \frac{y'}{y} \Rightarrow y''' y = y'' y'$$

16. Which of the following statements is/are true?

- (1) An integrating factor of the differential equation $\frac{dy}{dx} + \frac{2y}{x} = x^3 \sin x$ is x^2
- (2) An integrating factor of the differential equation $x dy - y dx = x^2 y^2 dy$ is $\frac{1}{x^2}$
- (3) An integrating factor of the differential equation $x \frac{dy}{dx} + xy \cot x = \operatorname{cosec} x$ is $\sin x$
- (4) An integrating factor of the differential equation $(y dx - x dy) = xy^3 (x dy + y dx)$ is $\frac{1}{y^2}$

Sol. Answer (1, 2, 3, 4)

(1) I.F. = $e^{\int \frac{2}{x} dx} = x^2 \rightarrow \text{true}$

(2) $x - y \frac{dx}{dy} = x^2 y^2$

$$y \frac{dx}{dy} - x = -x^2 y^2$$

$$\frac{1}{x^2} \frac{dx}{dy} - \frac{1}{xy} = -y$$

Let $\frac{1}{x} = t$

$$\Rightarrow -\frac{dt}{dy} - \frac{t}{y} = -y$$

$$\Rightarrow -\frac{1}{x^2} \frac{dx}{dy} = \frac{dt}{dy}$$

$$\Rightarrow \frac{dt}{dy} + \frac{t}{y} = y$$

$$\therefore \text{I.F.} = e^{\int \frac{1}{y} dy} = y$$

But we can also write the differential equation as $x dy - y dx = x^2 y^2 dy$

Multiplying by $\frac{1}{x^2}$, we get

$$\Rightarrow d\left(\frac{y}{x}\right) = d\left(\frac{y^3}{3}\right) \Rightarrow \therefore \text{Equation becomes integrable}$$

$$\therefore \frac{1}{x^2} \text{ is an I.F.} \rightarrow \text{True}$$

(3) $x \frac{dy}{dx} + xy \cot x = \operatorname{cosec} x$

$$\frac{dy}{dx} + y \cot x = \frac{\operatorname{cosec} x}{x}$$

$$\text{I.F.} = e^{\int \cot x dx} = \sin x \rightarrow \text{true.}$$

$$(4) (ydx - xdy) = xy^3(xdy + ydx)$$

Multiplying by $\frac{1}{y^2}$, we get

$$\rightarrow d\left(\frac{x}{y}\right) = xy d(xy) \therefore \text{Equation becomes integrable.}$$

$$\rightarrow \frac{1}{y^2} \text{ is an integrating factor.}$$

17. A curve passes through (1, 0) and satisfies the differential equation $(2x \cos y + 3x^2y) dx + (x^3 - x^2 \sin y - y) dy = 0$

$$(1) \text{ The equation of curve is } x^2 \cos y + x^3 y - y^2 = 1 \quad (2) \text{ The equation of curve is } x^2 \cos y + x^3 y - \frac{y^2}{2} = 1$$

$$(3) \text{ The equation of normal at (1, 0) is } y = 0 \quad (4) \text{ The equation of tangent at (1, 0) is } x = 1$$

Sol. Answer (2, 3, 4)

$$(2x \cos y dx + x^2(-\sin y) dy) + (3x^2y dx + x^3 dy) - y dy = 0$$

$$\Rightarrow d(x^2 \cos y) + d(x^3 y) - d\left(\frac{y^2}{2}\right) = 0$$

$$\Rightarrow x^2 \cos y + x^3 y - \frac{y^2}{2} = c$$

It passes through (1, 0)

$$\Rightarrow 1 + 0 - 0 = c \Rightarrow c = 1$$

$$\therefore \text{Equation of curve is } x^2 \cos y + x^3 y - \frac{y^2}{2} = 1$$

$$\frac{dy}{dx} = \frac{2x \cos y + 3x^2 y}{y + x^2 \sin y - x^3}$$

$$\therefore \left. \frac{dy}{dx} \right|_{(1,0)} = \frac{0+0-0}{2+0} = 0$$

$\therefore$ Slope of normal = 0.

$\therefore$ equation of normal at (1, 0) is $y = 0$

Also the tangent at (1, 0) is vertical

$\therefore$ its equation is $x = 1$

18. If $y(x)$ satisfies the differential equation $y' - y \tan x = 2x \sec x$ and $y(0) = 0$, then

[IIT-JEE 2012]

$$(1) \quad y\left(\frac{\pi}{4}\right) = \frac{\pi^2}{8\sqrt{2}}$$

$$(2) \quad y'\left(\frac{\pi}{4}\right) = \frac{\pi^2}{18}$$

$$(3) \quad y\left(\frac{\pi}{3}\right) = \frac{\pi^2}{9}$$

$$(4) \quad y'\left(\frac{\pi}{3}\right) = \frac{4\pi}{3} + \frac{2\pi^2}{3\sqrt{3}}$$

Sol. Answer (1, 4)

$$\frac{dy}{dx} - y \tan x = 2x \sec x$$

$$\cos x \frac{dy}{dx} - (\sin x)y = 2x$$

$$d(y \cos x) = d(x^2)$$

$$y \cos x = x^2 + c$$

$$y(0) = c = 0$$

$$y = x^2 \sec x$$

$$\text{also, } y' = 2x \sec x + x^2 \sec x \tan x$$

19. Let $y(x)$ be a solution of the differential equation $(1 + e^x)y' + ye^x = 1$. If $y(0) = 2$, then which of the following statement is(are) true? [JEE(Advanced)-2015]

$$(1) \quad y(-4) = 0$$

$$(2) \quad y(-2) = 0$$

$$(3) \quad y(x) \text{ has a critical point in the interval } (-1, 0)$$

$$(4) \quad y(x) \text{ has no critical point in the interval } (-1, 0)$$

Sol. Answer (1, 3)

$$(1 + e^x)dy + ye^x dx = dx$$

$$dy + d(e^x \cdot y) = dx$$

$$ye^x = x - y + c$$

$$\text{Now, } x = 0, y = 2$$

$$2 \cdot e^0 = 0 - 2 + c$$

$$\Rightarrow c = 4$$

$$\text{Now, } ye^x = x - y + 4$$

$$y = \frac{x+4}{e^x + 1}$$

$$y(-4) = 0$$

$$\text{Now, } \frac{dy}{dx} = \frac{(e^x + 1) - (x+4)e^x}{e^x + 1}$$

$$= \frac{e^x + 1 - xe^x - 4e^x}{e^x + 1} = \frac{1 - xe^x - 3e^x}{e^x + 1}$$

$$f'(0) = -ve$$

$$f'(-1) = \frac{1 + e - 3e^{-1}}{e^{-1} + 1} = +ve$$

$$\text{Critical point in } (-1, 0)$$

20. Consider the family of all circles whose centers lie on the straight line $y = x$. If this family of circles is represented by the differential equation $Py'' + Qy' + 1 = 0$, where P, Q are functions of x, y and y' (here $y' = \frac{dy}{dx}$, $y'' = \frac{d^2y}{dx^2}$), then which of the following statements is(are) true? [JEE(Advanced)-2015]

- (1) $P = y + x$ (2) $P = y - x$
 (3) $P + Q = 1 - x + y + y' + (y')^2$ (4) $P - Q = x + y - y' - (y')^2$

Sol. Answer (2, 3)

Let required circle be

$$x^2 + y^2 + 2g(x + y) + c = 0$$

Differentiation

$$2x + 2yy' + 2g(1 + y') = 0$$

$$\Rightarrow x + yy' + g(1 + y') = 0 \quad \dots(i)$$

Again differentiation

$$1 + yy'' + (y')^2 + g(y'') = 0 \quad \dots(ii)$$

Putting value of g in equation (ii),

$$1 + yy'' + (y')^2 - \left(\frac{x + yy'}{1 + y'} \right) y'' = 0$$

$$\text{So, } P = y - x, Q = 1 + y' + (y')^2$$

$$\text{i.e., } P + Q = 1 - x + y + y' + (y')^2$$

21. A solution curve of the differential equation $(x^2 + xy + 4x + 2y + 4) \frac{dy}{dx} - y^2 = 0$, $x > 0$, passes through the point (1, 3). Then the solution curve [JEE(Advanced)-2016]
- (1) Intersects $y = x + 2$ exactly at one point (2) Intersects $y = x + 2$ exactly at two points
 (3) Intersects $y = (x + 2)^2$ (4) Does NOT intersect $y = (x + 3)^2$

Sol. Answer (1, 4)

$$(x^2 + xy + 4x + 2y + 4) \frac{dy}{dx} = y^2$$

$$(x + 2)^2 + y(x + 2) = y^2 \frac{dx}{dy}$$

$$\frac{dx}{dy} = \frac{(x + 2)^2}{y^2} + \frac{x + 2}{y}$$

$$\frac{1}{(x + 2)^2} \frac{dx}{dy} - \frac{1}{y(x + 2)} = \frac{1}{y^2}$$

$$t = \frac{-1}{x + 2}$$

$$\frac{dt}{dy} + \frac{t}{y} = \frac{1}{y^2}$$

$$IF = e^{\int \frac{dy}{y}} = y$$

$$y \cdot t = \int \frac{y}{y^2} dy = \ln y + c$$

$$\Rightarrow \frac{-y}{x+2} = \ln y + c$$

It passes through (1, 3)

$$\Rightarrow c = -1 - \ln 3$$

$$\Rightarrow \ln y = 1 - \frac{y}{x+2} + \ln 3$$

On solving with $y = x + 2$

$$\ln(x+2) = \ln 3 \Rightarrow x = 1 \quad (\text{Exactly one point})$$

On solving with $y = (x+2)^2$

$$\begin{aligned} \ln(x+2)^2 &= 1 - (x+2) + \ln 3 \\ &= -1 + \ln 3 - x \end{aligned}$$

As graph of LHS is increasing & graph of RHS is decreasing

$$\text{and } \lim_{x \rightarrow 0^+} \text{LHS} > \lim_{x \rightarrow 0^+} \text{RHS}$$

So the curves do not intersect

$$\text{Also, } (x+3)^2 > (x+2)^2 \text{ for } x > 0$$

$$\text{So, } y = (x+3)^2 \text{ and } \ln y = 1 + \ln 3 - \frac{y}{x+2} \text{ also do not intersect}$$

22. Let $f: (0, \infty) \rightarrow \mathbb{R}$ be a differentiable function such that $f'(x) = 2 - \frac{f(x)}{x}$ for all $x \in (0, \infty)$ and $f(1) \neq 1$. Then

[JEE(Advanced)-2016]

$$(1) \lim_{x \rightarrow 0^+} f'\left(\frac{1}{x}\right) = 1$$

$$(2) \lim_{x \rightarrow 0^+} x f\left(\frac{1}{x}\right) = 2$$

$$(3) \lim_{x \rightarrow 0^+} x^2 f'(x) = 0$$

$$(4) |f(x)| \leq 2 \text{ for all } x \in (0, 2)$$

Sol. Answer (1)

$$f'(x) + \frac{f(x)}{x} = 2$$

$$I.F = e^{\int \frac{1}{x} dx} = e^{\log x} = x$$

$$f(x) \cdot x = \int 2x dx + c$$

$$f(x) \cdot x = x^2 + c$$

$$f(x) = x + \frac{c}{x} \quad \therefore f(1) \neq 1$$

$$1 + \frac{c}{1} \neq 1 \quad c \neq 0$$

$$f'(x) = 1 - \frac{c}{x^2}$$

$$\Rightarrow f'\left(\frac{1}{x}\right) = 1 - cx^2$$

$$(1) \lim_{x \rightarrow 0^+} f'\left(\frac{1}{x}\right) = \lim_{x \rightarrow 0^+} (1 - cx^2) = 1$$

$$(2) \lim_{x \rightarrow 0^+} x \cdot f\left(\frac{1}{x}\right) = \lim_{x \rightarrow 0^+} x \left(\frac{1}{x} + cx\right) = \lim_{x \rightarrow 0^+} (1 + cx^2) = 1$$

$$(3) \lim_{x \rightarrow 0^+} x^2 f'(x) = \lim_{x \rightarrow 0^+} x^2 \left(1 - \frac{c}{x^2}\right) = \lim_{x \rightarrow 0^+} (x^2 - c) = -c$$

$$(4) f(x) = x + \frac{c}{x}$$

23. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ be two non-constant differentiable functions. If $f(x) = (e^{f(x)-g(x)}) g'(x)$ for all $x \in \mathbb{R}$, and $f(1) = g(2) = 1$, then which of the following statement(s) is (are) TRUE? [JEE(Advanced)-2018]

$$(1) f(2) < 1 - \log_e 2 \quad (2) f(2) > 1 - \log_e 2 \quad (3) g(1) > 1 - \log_e 2 \quad (4) g(1) < 1 - \log_e 2$$

Sol. Answer (2, 3)

$$f(x) = (e^{f(x)-g(x)}) \cdot g'(x)$$

$$\Rightarrow \frac{f'(x)}{e^{f(x)}} = \frac{g'(x)}{e^{g(x)}}$$

$$\Rightarrow e^{-f(x)} = e^{-g(x)} + c$$

$$\text{Put } x = 1, e^{-1} = e^{-g(1)} + c$$

$$\text{Put } x = 2, e^{-f(2)} = e^{-1} + c$$

$$\Rightarrow \frac{1}{e} - \frac{1}{e^{f(2)}} = \frac{1}{e^{g(1)}} - \frac{1}{e}$$

$$\text{i.e., } e^{g(1)} = \frac{e^{f(2)+1}}{2e^{f(2)} - e} > 0$$

$$\Rightarrow e^{f(2)} > \frac{e}{2}$$

$$\therefore f(2) > 1 - \ln 2$$

$$\text{Also, } e^{f(2)} = \frac{2e^{g(1)} - e}{e^{1+g(1)}} > 0 \Rightarrow g(1) > 1 - \ln 2$$

24. Let $f : [0, \infty) \rightarrow \mathbb{R}$ be a continuous function such that $f(x) = 1 - 2x + \int_0^x e^{x-t} f(t) dt$ for all $x \in [0, \infty)$. Then, which of the following statement(s) is (are) TRUE? [JEE(Advanced)-2018]

- (1) The curve $y = f(x)$ passes through the point $(1, 2)$
 (2) The curve $y = f(x)$ passes through the point $(2, -1)$
 (3) The area of the region $\{(x, y) \in [0, 1] \times \mathbb{R} : f(x) \leq y \leq \sqrt{1-x^2}\}$ is $\frac{\pi-2}{4}$
 (4) The area of the region $\{(x, y) \in [0, 1] \times \mathbb{R} : f(x) \leq y \leq \sqrt{1-x^2}\}$ is $\frac{\pi-1}{4}$

Sol. Answer (2, 3)

$$\therefore f(x) = 1 - 2x + e^x \int_0^x e^{-t} f(t) dt \quad \dots(1)$$

On differentiating both sides and using eq. (1)

$$f'(x) = 2f(x) + 2x - 3$$

$$\therefore \frac{dy}{dx} - 2y = 2x - 3$$

using linear differential equation concept:

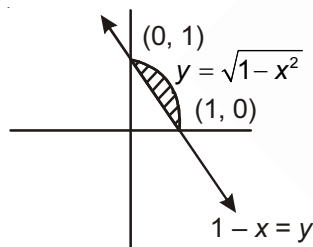
$$y = -x + 1 + c \cdot e^{-2x} \quad \dots(2)$$

$$\text{when } x = 0, y = 1 \Rightarrow c = 0$$

$$\therefore x + y = 1 \quad \dots(3)$$

It passes through $(2, -1)$

$$\text{Now } 1 - x \leq y \leq \sqrt{1-x^2}$$



$$\text{Required area} = \left(\frac{\pi}{4} - \frac{1}{2} \right) = \left(\frac{\pi-2}{4} \right) \text{ sq. units}$$

SECTION - C

Linked Comprehension Type Questions

Comprehension-I

Newton's law of cooling states that the rate of change of the temperature T of an object is proportional to the difference between T and the (constant) temperature τ of the surrounding medium, we can write it as

$$\frac{dT}{dt} = -k(T - \tau), k > 0 \text{ constant}$$

An cup of coffee is served at 185°F in a room where the temperature is 65°F . 2 minutes later the temperature of the coffee has dropped to 155°F

$$\log_e 3 = 1.09872, \log_e \frac{3}{4} = -0.2877$$

1. The temperature of any object at $t = 2$ is

(1) $\tau e^{-k} + [T(0)] e^{-2K}$ (2) $\tau e^k + [T(0) - \tau] e^{-2K}$ (3) $\tau + [T(0) - \tau] e^{-2K}$ (4) $\tau + 2[T(0) + \tau] e^{-K}$

Sol. Answer (3)

$$\frac{dT}{dt} = -k(T - \tau), k > 0$$

$$\frac{dT}{T - \tau} = -k dt \quad \dots(i)$$

Let the temperature of object at $t = 0$ be $T(0)$ and at $t = t_1$ it becomes T_1 . Now integrating the equation (i).

$$\int_{T(0)}^{T_1} \frac{dT}{T - \tau} = - \int_0^{t_1} k dt$$

$$\Rightarrow [\ln(T - \tau)]_{T(0)}^{T_1} = -k t_1$$

$$\Rightarrow \ln\left(\frac{T_1 - \tau}{T(0) - \tau}\right) = -k t_1 \quad \dots(ii)$$

or $T_1 - \tau = e^{-k t_1} (T(0) - \tau)$

$$T_1 = \tau + e^{-k t_1} (T(0) - \tau)$$

at $t_1 = 2$

$$T_1 = \tau + e^{-2k} (T(0) - \tau)$$

Option (3) is correct.

2. Time required for coffee to have 105°F temperature is

(1) 6 minute (2) 6.43 minute (3) 7.23 minute (4) 7.63 minute

Sol. Answer (4)

For coffee $T(0) = 185^\circ\text{F}$, $T_1 = 155^\circ\text{F}$ if $t_1 = 2$ minutes.

Let time required to have the temperature 105°F is t_2 .

Then we may use equation (ii) that is

$$\ln\left(\frac{T_1 - \tau}{T(0) - \tau}\right) = -k t_1 \quad \dots(iii)$$

$$\ln\left(\frac{T_2 - \tau}{T(0) - \tau}\right) = -k t_2 \quad \dots(iv)$$

Dividing (iii) and (iv)

$$\frac{\ln\left(\frac{T_1 - \tau}{T(0) - \tau}\right)}{\ln\left(\frac{T_2 - \tau}{T(0) - \tau}\right)} = \frac{-k t_1}{-k t_2} = \frac{t_1}{t_2}$$

Putting the values

$$\frac{\ln\left(\frac{155-65}{185-65}\right)}{\ln\left(\frac{105-65}{185-65}\right)} = \frac{2}{t_2}$$

$$\Rightarrow \frac{\ln\left(\frac{90}{120}\right)}{\ln\left(\frac{40}{120}\right)} = \frac{2}{t^2} \Rightarrow \frac{\ln\left(\frac{3}{4}\right)}{\ln\left(\frac{1}{3}\right)} = \frac{2}{t^2}$$

$$\Rightarrow \frac{\ln\left(\frac{4}{3}\right)}{\ln 3} = \frac{2}{t^2}$$

$$\frac{.2925}{1.09872} = \frac{2}{t_2}$$

$$t_2 = \frac{1.09872}{.1462} = 7.63 \text{ min}$$

Hence option (4) is correct.

3. Temperature of coffee at time t is given by

(1) $65 + 120e^{-kt}$

(2) $75 + 110e^{-kt}$

(3) $65 + 140e^{-2kt}$

(4) $75 + 140e^{-kt}$

Sol. Answer (1)

Temperature of coffee at any time ' t ' is given by

$$T_1 - \tau = e^{-kt} (T(0) - \tau)$$

at $t_1 = t$, $T = T_1$, and $T(0) = 185^\circ\text{F}$, $T = 65$

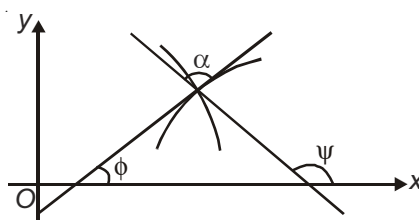
$$\Rightarrow T = 65 + 120e^{-kt}$$

Option (1) is correct.

Comprehension-II

Isogonal Trajectories : Let the trajectories cut the curve of given family at an angle α where $\tan \alpha = k$.

The slope $\frac{dy}{dx} = \tan \phi$ (of the tangent to a member of the family and the slope $\frac{dy_T}{dx} = \tan \psi$ to the isogonal trajectory are connected by the relationship



$$\tan \phi = \tan(\Psi - \alpha) = \frac{\tan \Psi - \tan \alpha}{1 + \tan \alpha \tan \Psi}$$

$$i.e., \frac{dy}{dx} = \frac{\frac{dy_T}{dx} - k}{k \frac{dy_T}{dx} + 1}$$

Substituting this expression into equation. (I') and dropping the subscript T , we obtain the differential equation of isogonal trajectories.

1. The isogonal trajectories of the family of straight lines $y = cx$, that cut the lines of the given family at an angle α , the tangent of which equal k is

$$(1) \quad x^2 + y^2 = c^2$$

$$(2) \quad y^2 = 4cx$$

$$(3) \quad \frac{x^2}{c^2} + \frac{y^2}{2-c^2} = 1$$

$$(4) \quad (x^2 + y^2)^{\frac{1}{2}} = ce^{\frac{\tan^{-1}\left(\frac{y}{x}\right)}{k}}$$

Sol. Answer (4)

The given curve is

$$\frac{dy}{dx} = c$$

$$\text{or } \frac{dy}{dx} = \frac{y}{x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{k + \frac{y}{x}}{1 - k \frac{y}{x}}$$

Integrating this homogeneous equation we get.

$$\ln(\sqrt{x^2 + y^2}) = \frac{1}{k} \tan^{-1}\left(\frac{y}{x}\right) + \ln c$$

2. The isogonal trajectories of a family of straight lines $y = c$, that cuts the given family at angle α , the tangent of which is k , is

$$(1) \quad y = kx$$

$$(2) \quad y = k \tan \alpha x$$

$$(3) \quad y = k \cot 2x$$

$$(4) \quad y = cx$$

Sol. Answer (1)

$$y = c$$

$$\Rightarrow \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{k+0}{1-0}$$

$$\Rightarrow y = kx$$

Comprehension-III

A family of curves is such that the slope of normal at any point (x, y) is $2(1 - y)$.

1. If $y = f(x)$ is a member of this family passing through $(-1, 2)$ then its equation is

- (1) $y^2 - 2y - x - 1 = 0$ (2) $y^2 + 2y + x - 7 = 0$ (3) $y^2 - 2y + x + 1 = 0$ (4) $y^2 + 2y - x - 9 = 0$

Sol. Answer (1)

$$\frac{dy}{dx} = -\frac{1}{2(1-y)}$$

$$\Rightarrow 2(y-1)dy = dx$$

$$(y-1)^2 = x + c$$

If it passes through $(-1, 2)$, then

$$1 = -1 + c \Rightarrow c = 2$$

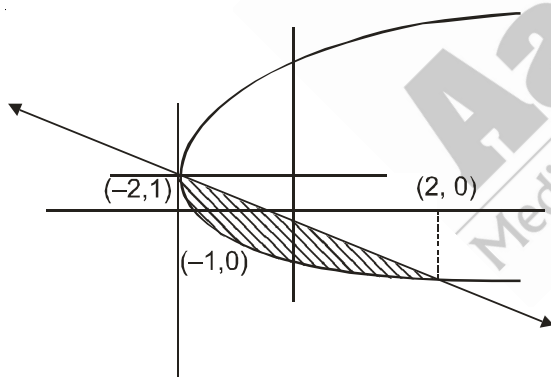
$\therefore$ Equation of curve is $(y-1)^2 = x+2$

$$\Rightarrow y^2 - 2y - x - 1 = 0 \Rightarrow \text{Option (1) is correct.}$$

2. The area bounded by the curve $y = f(x)$ of question number 1 and the line $x + 2y = 0$ is

- (1) $\frac{10}{3}$ sq. units (2) $\frac{4}{3}$ sq. units
(3) $\frac{28}{3}$ sq. units (4) $\frac{16}{3}$ sq. units

Sol. Answer (2)



$$\frac{dy}{dx} = -\frac{1}{2(1-y)}$$

$$\Rightarrow 2(y-1)dy = dx$$

$$(y-1)^2 = x + c$$

Required Area = ?

$$(y-1)^2 = x+2, \quad x+2y=0$$

$$\left(-\frac{x}{2}-1\right)^2 = x+2$$

$$(x+2)^2 - 4(x+2) = 0$$

$$(x+2)(x-2) = 0 \Rightarrow x = \pm 2$$

Required area

$$= \int_{-2}^2 (y_L - y_P) dx$$

$$= \int_{-2}^2 \left\{ -\frac{x}{2} - (-\sqrt{x+2} + 1) \right\} dx$$

$$= 0 + \int_{-2}^2 \sqrt{x+2} dx - 2 \int_0^2 dx$$

$$= \frac{2}{3} (x+2)^{3/2} \Big|_{-2}^2 - 2x \Big|_0^2$$

$$= \frac{2}{3} (8-0) - 2 \times 2$$

$$= \frac{4}{3} \text{ sq. units}$$

3. The orthogonal trajectories of the given family of curves is

(1) $y = ke^{-2x} - 1$

(2) $y = ke^{2x} + 1$

(3) $y = ke^{+2x} - 1$

(4) $y = ke^{-2x} + 1$

Sol. Answer (4)

$$\frac{dy}{dx} = -\frac{1}{2(1-y)}$$

$$\Rightarrow 2(y-1)dy = dx$$

$$(y-1)^2 = x + c$$

$$\therefore \frac{dy}{dx} = \frac{1}{2(y-1)}$$

$\therefore$ D.E. of orthogonal trajectories is

$$-\frac{dx}{dy} = \frac{1}{2(y-1)}$$

$$\Rightarrow \frac{dy}{y-1} = -2dx$$

$$\Rightarrow \ln|y-1| = -2x + \ln k$$

$$\therefore y-1 = ke^{-2x}$$

$$\Rightarrow y = 1 + ke^{-2x}$$

Comprehension-IV

A tangent to a curve at $P(x, y)$ intersects x -axis and y -axis at A and B respectively. Let the point of contact divides AB in the ratio $y^2 : x^2$.

1. The differential equation of family of curves is

(1) $x^2 + y^2 = c^2$ (2) $x^2 + y^2 - 2x = c$ (3) $x^2 + y^2 = cx^2y^2$ (4) $xy = c$

Sol. Answer (1)

$$Y - y = \frac{dy}{dx}(X - x)$$

$$A: \left(x - \frac{y}{dy/dx}, 0 \right); B: \left(0, y - x \frac{dy}{dx} \right)$$

$$P(x, y) = \left(\frac{0 + x^2 \left(x - \frac{y}{dy/dx} \right)}{x^2 + y^2}, \frac{y^2 \left(y - x \frac{dy}{dx} \right) + 0}{x^2 + y^2} \right)$$

$$\Rightarrow x(x^2 + y^2) = x^3 - \frac{yx^2}{dy/dx}$$

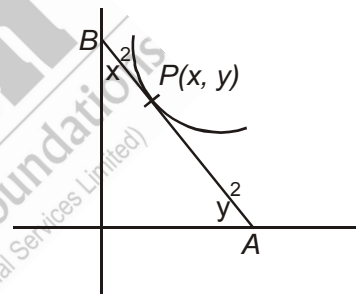
$$\Rightarrow xy^2 = -x^2 y \frac{dx}{dy}$$

$$\Rightarrow y^3 + yx^2 = y^3 - xy^2 \frac{dy}{dx}$$

$$\Rightarrow x = -y \frac{dy}{dx}$$

$$y dy + x dx = 0$$

$$x^2 + y^2 = c^2$$



2. If a member of this family passes through (3, 4), then its equation is

(1) $x^2 + y^2 = 25$ (2) $x^2 + y^2 - 2x = 19$ (3) $x^2 + y^2 = 25x^2y^2$ (4) $x^2 + y^2 = 7$

Sol. Answer (1)

$$Y - y = \frac{dy}{dx}(X - x)$$

$$A: \left(x - \frac{y}{dy/dx}, 0 \right); B: \left(0, y - x \frac{dy}{dx} \right)$$

$$P(x, y) = \left(\frac{0 + x^2 \left(x - \frac{y}{dy/dx} \right)}{x^2 + y^2}, \frac{y^2 \left(y - x \frac{dy}{dx} \right) + 0}{x^2 + y^2} \right)$$

$$\Rightarrow x(x^2 + y^2) = x^3 - \frac{yx^2}{dy/dx}$$

$$\Rightarrow xy^2 = -x^2 y \frac{dx}{dy}$$

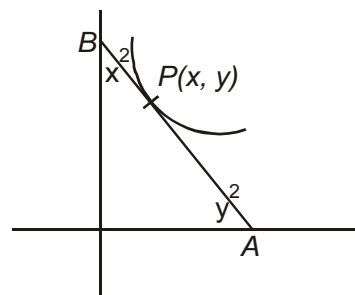
$$\Rightarrow y^3 + yx^2 = y^3 - xy^2 \frac{dy}{dx}$$

$$\Rightarrow x = -y \frac{dy}{dx}$$

It passes through (3, 4)

$$\Rightarrow c^2 = 25$$

$$\therefore x^2 + y^2 = 25$$



3. The area of the curve in question no. 2 is

(1) $\left(\frac{25}{4} + \frac{2\pi}{3}\right)$

(2) $2(\pi + 4)$

(3) 25π

(4) $\left(16\pi + \frac{4}{5}\right)$

Sol. Answer (3)

$$Y - y = \frac{dy}{dx}(X - x)$$

$$A: \left(x - \frac{y}{dy/dx}, 0\right); B: \left(0, y - x \frac{dy}{dx}\right)$$

$$P(x, y) = \left(\frac{0 + x^2 \left(x - \frac{y}{dy/dx}\right)}{x^2 + y^2}, \frac{y^2 \left(y - x \frac{dy}{dx}\right) + 0}{x^2 + y^2} \right)$$

$$\Rightarrow x(x^2 + y^2) = x^3 - \frac{yx^2}{dy/dx}$$

$$\Rightarrow xy^2 = -x^2 y \frac{dx}{dy}$$

$$\Rightarrow y^3 + yx^2 = y^3 - xy^2 \frac{dy}{dx}$$

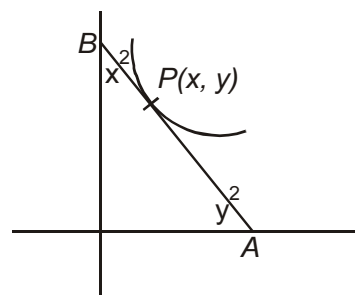
$$\Rightarrow x = -y \frac{dy}{dx}$$

$$\text{Area of circle} = \pi r^2$$

$$= 25\pi \text{ sq. units}$$

$$y - 1 = ke^{-2x}$$

$$y = 1 + ke^{-2k}$$



SECTION - D

Assertion-Reason Type Questions

1. STATEMENT-1 : To find complete solution of a second order differential equation we need two different conditions.

and

STATEMENT-2 : An n^{th} order differential equation has n independent parameters.

Sol. Answer (1)

Statement-1 is true and statement-2 is true also statement-2 is correct explanation of statement-1.

2. STATEMENT-1 : The orthogonal trajectory of a family of circles touching x-axis at origin and whose centre the on y-axis is self orthogonal.

and

STATEMENT-2 : In order to find the orthogonal trajectory of a family of curves we put $-\frac{dx}{dy}$ in place of $\frac{dy}{dx}$ in the differential equation of the given family of curves.

Sol. Answer (4)

The equation of a family of circles touching x-axis at origin and whose centre lies on y-axis may be written as

$x^2 + y^2 - ay = 0$, where a is a parameter

$$\Rightarrow 2x + 2y \frac{dy}{dx} - a \frac{dy}{dx} = 0$$

$$\Rightarrow 2x + 2y - \frac{x^2 + y^2}{y} \frac{dy}{dx} = 0$$

$$\Rightarrow 2x + \frac{y^2 - x^2}{y} \frac{dy}{dx} = 0$$

For orthogonal trajectory,

$$2x + \frac{y^2 - x^2}{y} \left(-\frac{dx}{dy} \right) = 0$$

$$\Rightarrow \frac{2xy}{y^2 - x^2} = \frac{dx}{dy} \quad \Rightarrow \frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$$

Let us put $y = vx$

$$\Rightarrow u + x \frac{dy}{dx} = \frac{v^2 - 1}{2v}$$

$$\Rightarrow \frac{dx}{x} + \frac{2v}{1+v^2} dv = 0$$

$$\Rightarrow \ln x(1+v^2) = \ln k$$

$$\Rightarrow x^2 + y^2 = kx, \text{ which represents a family of circles touching y-axis at origin.}$$

$\Rightarrow$ Statement-1 is false and Statement-2 is true.

3. STATEMENT-1 : If the length of subtangent and subnormal at point (x, y) on $y = f(x)$ are 9 and 4 then x is equal to + 6.

and

STATEMENT-2 : Product of subtangent and subnormal is square of the ordinate of the point.

Sol. Answer (4)

$$\left| \frac{y_1}{\frac{dy}{dx}} \right| = 9 \text{ and } \left| y_1 \frac{dy}{dx} \right| = 4 \quad |y_1|^2 = 36 \Rightarrow y_1 = \pm 6$$

Product of sub-tangent and subnormal is y_1^2 (statement 1 is false.)

4. STATEMENT-1 : The differential equation of all non-horizontal lines in a plane is $\frac{d^2y}{dx^2} = 0$

and

STATEMENT-2 : The general equation of all non-horizontal line in xy plane is $ax + by = 1$, $a \neq 0$

Sol. Answer (4)

Equation of non-horizontal lines in a plane is $ax + by = 1$ ($a \neq 0$) or $a \frac{dx}{dy} + b = 0$ ($a \neq 0$ and $b \in R$)

$$\text{or } a \frac{d^2x}{dy^2} = 0 \text{ or } \frac{d^2x}{dy^2} = 0$$

5. STATEMENT-1 : The differential equation whose general solution is $y = c_1x + \frac{c_2}{x}$ for all values of c_1 and c_2 is linear equation.

and

STATEMENT-2 : The equation $y = c_1x + \frac{c_2}{x}$ has two arbitrary constants, so the corresponding differential equation is second order.

Sol. Answer (2)

$$y = c_1x + \frac{c_2}{x} \Rightarrow \frac{dy}{dx} = c_1 - \frac{c_2}{x^2} \Rightarrow \frac{d^2y}{dx^2} = \frac{2c_2}{x^3}$$

Eliminating c_1 and c_2 from above three equations,

$$\text{we get, } \frac{d^2y}{dx^2} + \frac{1}{x} \frac{dy}{dx} - \frac{y}{x^2} = 0 \text{ which is a linear equation.}$$

6. STATEMENT-1 : The differential equation $\frac{d^2y}{dx^2} + \cos x \frac{dy}{dx} + (x^3 + 7)y = e^x$ is a linear equation.

and

STATEMENT-2 : Every first degree equation is a linear equation.

Sol. Answer (3)

A linear differential equation is an equation in which the dependent variable and its derivatives appear only in first degree

7. STATEMENT-1 : The differential equation $\frac{dy}{dx} = \frac{2xy}{x^2 + y^2}$ Can't be solved by the substitution $x = vy$.

and

STATEMENT-2 : When the differential equation is homogeneous of first order and first degree, then the substitution that solves the equation is $y = vx$.

Sol. Answer (4)

The differential equation can be put in the form $\frac{dx}{dy} = \frac{x^2 + y^2}{2xy}$ and it being homogeneous of first order and first degree with roles of x and y interchanged, can be solved by the substitution $x = vy$. Thus Statement-1 is false. Again Statement-2 is true. The answer then is (4).

8. STATEMENT-1 : Solution of the differential equation $xdy - y dx = y dy$ is $ye^{x/y} = c$.

and

STATEMENT-2 : Given differential equation can be re-written as $d\left(\frac{x}{y}\right) = -\frac{dy}{y}$.

Sol. Answer (1)

$$\frac{xdy - ydx}{y} = dy \Rightarrow \frac{ydx - xdy}{y^2} = -\frac{dy}{y}$$

$$\Rightarrow d\left(\frac{x}{y}\right) = -\frac{dy}{y} \Rightarrow \frac{x}{y} = -\log y + \log c = \frac{x}{y} = \log \frac{c}{y}$$

$$\Rightarrow y e^{\frac{x}{y}} = c$$

9. STATEMENT-1 : The differential equation of all circles in a plane can be of order 3.

and

STATEMENT-2 : General equation of a circle in plane has three independent constant parameters.

Sol. Answer (1)

General equation of the circle is $x^2 + y^2 + 2gx + 2fy + c$ has 3 independent constant parameters. Hence order is 3.

10. Let a solution $y = y(x)$ of the differential equation $x\sqrt{x^2 - 1} dy - y\sqrt{y^2 - 1} dx = 0$ satisfy $y(2) = \frac{2}{\sqrt{3}}$

$$\text{STATEMENT-1 : } y(x) = \sec\left(\sec^{-1} x - \frac{\pi}{6}\right)$$

and

$$\text{STATEMENT-2 : } y(x) \text{ is given by } \frac{1}{y} = \frac{2\sqrt{3}}{x} - \sqrt{1 - \frac{1}{x^2}}$$

[IIT-JEE 2008]

Sol. Answer (3)

$$\frac{dy}{dx} = \frac{y\sqrt{y^2-1}}{x\sqrt{x^2-1}}$$

$$\int \frac{dy}{y\sqrt{y^2-1}} = \int \frac{dx}{x\sqrt{x^2-1}}$$

$$\sec^{-1} y = \sec^{-1} x + c$$

$$x = 2$$

$$y = \frac{2}{\sqrt{3}}$$

$$\sec^{-1} \frac{2}{\sqrt{3}} = \sec^{-1} 2 + c$$

$$\frac{\pi}{6} = \frac{\pi}{3} + c$$

$$c = -\frac{\pi}{6}$$

$$y = \sec(\sec^{-1} x - \pi/6)$$

$$= \cos \left[\cos^{-1} \frac{1}{x} - \cos^{-1} \frac{\sqrt{3}}{2} \right] = \omega$$

$$= \cos \left[\cos^{-1} \left(\frac{\sqrt{3}}{2x} + \sqrt{1 - \frac{1}{x^2}} \sqrt{1 - \frac{3}{4}} \right) \right]$$

$$\frac{1}{y} = \frac{\sqrt{3}}{2x} + \frac{1}{2} \sqrt{1 - \frac{1}{x^2}}$$

SECTION - E**Matrix-Match Type Questions**

1. The solution of differential equations

Column-I

(A) $\left(x \cos \frac{y}{x} + y \sin \frac{y}{x} \right) y dx + \left(x \cos \frac{y}{x} - y \sin \frac{y}{x} \right) x dy = 0$

(B) $y_1 = \frac{x}{\cos y} - \tan y$

(C) $y - y_1 \cos x = y^2 \cos x (1 - \sin x)$

(D) $2yy_1 = e^{\frac{x^2+y^2}{x}} + \frac{x^2+y^2}{x} - 2x$

Column-II

(p) $y = \frac{\tan x + \sec x}{c + \sin x}$

(q) $\sin y = x - 1 + ce^{-x}$

(r) $\ln |cx| = -e^{\frac{x^2+y^2}{x}}$

(s) $xy \cos \frac{y}{x} = C$

Sol. Answer A(s), B(q), C(p), D(r)

(A) Let us put $\frac{y}{x} = v$ so that $y = vx$ and $\frac{dy}{dx} = v + x \frac{dv}{dx}$

The given differential equation reduces to

$$\frac{dy}{dx} = \frac{y}{x} \cdot \frac{x \cos \frac{y}{x} + y \sin \frac{y}{x}}{y \sin \frac{y}{x} - x \cos \frac{y}{x}}$$

$$\Rightarrow v + x \frac{dv}{dx} = v \cdot \frac{\cos v + v \sin v}{v \sin v - \cos v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v \cos v + v^2 \sin v - v^2 \sin v + v \cos v}{v \sin v - \cos v}$$

$$\Rightarrow \frac{v \sin v - \cos v}{v \cos v} dv = \frac{2dx}{x}$$

$$\Rightarrow 2 \frac{dx}{x} + \frac{\cos v - v \sin v}{v \cos v} dv = 0$$

$$\Rightarrow 2 \ln x + \ln v \cos v = \ln c$$

$$\Rightarrow \frac{x^2 y}{x} \cos \frac{y}{x} = c$$

$$\Rightarrow xy \cos \frac{y}{x} = c$$

(B) We have, $y_1 = \frac{x}{\cos y} - \tan y$

$$\Rightarrow \cos y \frac{dy}{dx} = x - \sin y$$

$$\Rightarrow \frac{d}{dx}(\sin y) + \sin y = x$$

Which is linear differential equation

$$\text{Integrating factor} = e^{\int dx} = e^x$$

$$\text{The solution is given by } e^x \sin y = \int x e^x dx = (x-1)e^x + c$$

$$\Rightarrow \sin y = (x-1) + ce^{-x}$$

(C) We have, $y - y_1 \cos x = y^2 \cos x(1 - \sin x)$

$$\Rightarrow \frac{dy}{dx} \cos x = y - y^2 \cos x(1 - \sin x)$$

$$\Rightarrow \frac{1}{y^2} \frac{dy}{dx} - \frac{1}{y} \sec x = -(1 - \sin x)$$

$$\Rightarrow \frac{d}{dx} \left(\frac{1}{y} \right) + (\sec x) \frac{1}{y} = (1 - \sin x)$$

Which is linear differential equation whose integrating factor = $e^{\int \sec x dx} = e^{\ln(\sec x + \tan x)} = \sec x + \tan x$

The solution of the differential equation is

$$\frac{1}{y}(\sec x + \tan x) = \int \frac{(1 - \sin x)(1 + \sin x)}{\cos x} dx = \sin x + c$$

$$\Rightarrow y = \frac{\sec x + \tan x}{c + \sin x}$$

(D) The given differential equation is $2y \frac{dy}{dx} - \frac{x^2 + y^2}{x} + 2x = e^{\frac{x^2 + y^2}{x}}$

Let us put $\frac{x^2 + y^2}{x} = z$

So that $2x + 2y \frac{dy}{dx} = z + x \frac{dz}{dx}$

$$\Rightarrow 2y \frac{dy}{dx} - \frac{x^2 + y^2}{x} + 2x = x \frac{dz}{dx} = e^z$$

$$\Rightarrow e^{-z} dz = \frac{dx}{x} \Rightarrow \frac{e^{-z}}{-1} = \ln |cx|$$

$$\Rightarrow \ln |cx| = -e^{-\frac{x^2 + y^2}{x}}$$

2. Match the following

Column-I

(A) $\frac{dy}{dx} + \frac{2y}{x} = 0$, $y(1) = 2$, $y(x_0) = 8$ then x_0 is

(B) $(x^2 + y^2)dy = xydx$ and $y(1) = 1$ and $y(x_0) = e$, then x_0 is

(C) If $y(u)$ is solution of $(u+1)\frac{dy}{du} - uy = 1$, $y(0) = -1$, then $y(1)$ is

(D) $x dy = y(dx + ydy)$, $y > 0$, $y(1) = 1$ and $y(x_0) = -3$, then x_0 is

Column-II

(p) $\frac{1}{2}$

(q) Rational number

(r) $\sqrt{3}e$

(s) -15

(t) $-\frac{1}{2}$

Sol. Answer A(p, q, t), B(r), C(q, t), D(q, s)

(A) $\int \frac{dy}{y} + 2 \int \frac{dy}{x} = 0 \Rightarrow \ln yx^2 = \ln c$

$$y = \frac{c}{x^2}$$

$$y(1) = 2 \Rightarrow 2 = \frac{c}{1^2} \Rightarrow c = 2$$

$$y = \frac{2}{x^2}$$

$$y(x_0) = 8 = \frac{2}{x_0^2} \Rightarrow x_0^2 = \frac{1}{4} \Rightarrow x_0 = \pm \frac{1}{2}$$

$$(B) \frac{dy}{dx} = \frac{v}{1+v^2} \quad \left(v = \frac{y}{x} \right)$$

$$v + x \cdot \frac{dv}{dx} = \frac{v}{1+v^2}$$

$$\Rightarrow \int \frac{dx}{x} + \int \left(\frac{1+v^2}{v^3} \right) dv = 0$$

$$\Rightarrow \ln x + \ln v - \frac{1}{2v^2} = c$$

$$\Rightarrow \ln xv = c + \frac{x^2}{2y}$$

$$\Rightarrow \ln y = \frac{x^2}{2y^2} + c$$

$$\ln 1 = \frac{1}{2} + c \Rightarrow c = -\frac{1}{2}$$

$$\ln y = \frac{x^2}{2y^2} - \frac{1}{2}$$

$$\ln e = \frac{x_0^2}{2e^2} - \frac{1}{2} \Rightarrow \frac{x_0^2}{2e^2} = \frac{3}{2} \Rightarrow x_0^2 = 3e^2 \Rightarrow x_0 = \sqrt{3}e \quad (y > 0)$$

$$(C) \frac{dy}{du} + y \left(-\frac{u}{u+1} \right) = \frac{1}{u+1} \quad \text{I.F} = e^{-\int \frac{u}{u+1} du} = e^{-u} (u+1)$$

$$\Rightarrow y \cdot e^{-u} (u+1) = \int e^{-u} du = -e^{-u} + c$$

$$\text{At } u = 0, y = -1 \Rightarrow c = 0 \quad \therefore y = -\frac{1}{u+1}$$

$$u = 1, \quad y(1) = -\frac{1}{1+1} = -\frac{1}{2}$$

$$(D) \frac{(dx)y - x dy}{y^2} + dy = 0$$

$$\int \frac{(dx)y - x dy}{y^2} + \int dy = 0$$

$$\int 1 \cdot d\left(\frac{x}{y}\right) + \int 1 \cdot dy = 0$$

$$\frac{x}{y} + y = c$$

$$\text{At } x = 1, y = 1 \Rightarrow c = 2$$

$$\therefore \frac{x}{y} + y = 2 \quad \text{At } x = x_0, y = -3$$

$$-\frac{x_0}{3} - 3 = 2$$

$$x_0 = -3(5) = -15$$

3. Match the following for arbitrary constants $a, b, c, d, c_1, c_2, c_3, c_4$

Column-I**Column-II**

- | | |
|---|---------------------------------|
| (A) $y = cx + c^2 - 3c^{\frac{3}{2}} + 2$ | (p) Order 2, degree not defined |
| (B) $\frac{d^2y}{dx^2} = \sin\left(\frac{dy}{dx}\right) + xy$ | (q) Order 1, degree 4 |
| (C) $y = (c_1 + c_2)\cos(x + c_3) - c_4 e^{x+c_5}$ | (r) Order 2, degree 2 |
| (D) $y = a\sin^2 x + b\cos^2 x + c\sin 2x + d\cos 2x$ | (s) Order 3, degree 1 |
| | (t) Order 3, degree infinite |

Sol. Answer A(q), B(p), C(s), D(s)

- (A) Differentiating the equation, we get $\frac{dy}{dx} = c$, eliminatory c we get $y = x \frac{dy}{dx} + \left(\frac{dy}{dx}\right)^2 - 3\left(\frac{dy}{dx}\right)^{\frac{3}{2}} + 2$ clearly

its order is 1 and removing fractional power, it will result in to 4th degree.

- (B) Order is 2 but degree cannot be determine because the equation is not expressible as polynomial.

- (C) The equation can be rewritten as $y = a\cos(x + b) + ce^x$.

- (D) $y = a\left(\frac{a - \cos 2x}{2}\right) + b\left(\frac{1 + \cos 2x}{2}\right) + 2\sin 2x + d\cos 2x$
 $= A + B\sin 2x + c\sin 2x$

4. Match the statements/expressions in Column I with the open intervals in Column II.

[IIT-JEE 2009]

Column I**Column II**

- (A) Interval contained in the domain of definition of non-zero solutions of the differential equation $(x - 3)^2 y' + y = 0$

(p) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

- (B) Interval containing the value of the integral

(q) $\left(0, \frac{\pi}{2}\right)$

$$\int_1^5 (x-1)(x-2)(x-3)(x-4)(x-5)dx$$

- (C) Interval in which at least one of the points of local maximum of $\cos^2 x + \sin x$ lies

(r) $\left(\frac{\pi}{8}, \frac{5\pi}{4}\right)$

- (D) Interval in which $\tan^{-1}(\sin x + \cos x)$ is increasing

(s) $\left(0, \frac{\pi}{8}\right)$

(t) $(-\pi, \pi)$

Sol. Answer A(p, q, s), B(p, t), C(p, q, r, t), D(s)

We have,

(A) $\frac{y'}{y} = -\frac{1}{(x-3)^2}$

$$\Rightarrow \int \frac{dy}{y} = \int -\frac{dx}{(x-3)^2}$$

$$\Rightarrow \log |y| = \frac{1}{x-3} + c$$

$$\Rightarrow x \neq 3$$

$$(B) I = \int_1^5 (x-1)(x-2)(x-3)(x-4)(x-5)dx$$

$$I = \int_1^5 (6-x-1)(6-x-2)(6-x-3)(6-x-4)(6-x-5)dx$$

$$I = \int_1^5 (5-x)(4-x)(3-x)(2-x)(1-x)dx$$

$$I = \int_1^5 -(x-5)(x-4)(x-3)(x-2)(x-1)dx$$

$$\Rightarrow I = -I$$

$$\Rightarrow 2I = 0$$

$$\Rightarrow I = 0$$

$$(C) y = 1 - \sin^2 x + \sin x$$

$$y = 1 - \left(\sin^2 x - \sin x + \frac{1}{4} - \frac{1}{4} \right)$$

$$y = \frac{5}{4} - \left(\sin x - \frac{1}{2} \right)^2$$

$$y \text{ is maximum at } \sin x = \frac{1}{2}$$

$$\Rightarrow x = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\Rightarrow x \in \left(-\frac{\pi}{2}, \frac{\pi}{2} \right) \cup \left(0, \frac{\pi}{2} \right) \cup (-\pi, \pi) \cup \left(\frac{\pi}{8}, \frac{5\pi}{4} \right)$$

$$(D) y = \tan^{-1}(\sin x + \cos x)$$

$$\frac{dy}{dx} = \frac{\cos x - \sin x}{1 + (\sin x + \cos x)^2}$$

$$\text{For increasing function } \frac{dy}{dx} > 0$$

$$\Rightarrow \cos x - \sin x > 0$$

$$\Rightarrow \cos x > \sin x$$

$$\Rightarrow x \in \left(0, \frac{\pi}{8} \right)$$

SECTION - F

Integer Answer Type Questions

1. If $x \sin\left(\frac{y}{x}\right) dy = \left(y \sin\left(\frac{y}{x}\right) - x\right) dx$ and $y(1) = \frac{\pi}{2}$ then the value of $\cos\left(\frac{y}{e^7}\right)$ is _____.

Sol. Answer (7)

$$\frac{dy}{dx} = \frac{\frac{y}{x} \sin \frac{y}{x} - 1}{\sin \frac{y}{x}} \Rightarrow v + x \cdot \frac{dv}{dx} = \frac{v \sin v - 1}{\sin v} = v - \frac{1}{\sin v}$$

$$\Rightarrow v + x \cdot \frac{dv}{dx} = v - \frac{1}{\sin v} \Rightarrow \int \sin v \, dv + \int \frac{dx}{x} = 0$$

$$\Rightarrow \ln x - \cos v = c$$

$$\Rightarrow \ln x = \cos \frac{y}{x} + c, \text{ At } x = 1, y = \frac{\pi}{2}, \text{ therefore } c = 0$$

$$\text{Hence } \ln x = \cos \frac{y}{x}$$

2. If $xy = ae^x + be^{-x}$ satisfies the equation $Axy'' + By' = xy$, then $|A - B|$ is _____.

Sol. Answer (1)

$$xy = ae^x + be^{-x} \quad \dots(1)$$

$$\Rightarrow y + xy' = ae^x - be^{-x} \quad \dots(2)$$

$$\Rightarrow xy' + xy'' = ae^x + be^{-x} \quad \dots(3)$$

From (1) and (3), we get

$$xy'' + 2y' = xy$$

$$\Rightarrow |A - B| = 1$$

3. If $\sec^2 y \frac{dy}{dx} + 2x \tan y = x^3$ satisfies $\tan y = ce^{-x^4} + B\{x^2 - 1\}$, then $(AB)^2$ equals _____.

Sol. Answer (1)

$$\text{Let } \tan y = z \Rightarrow \sec^2 y \frac{dy}{dx} = \frac{dz}{dx}$$

$$\therefore \frac{dz}{dx} + (2x)z = x^3 \Rightarrow P = 2x, Q = x^3$$

$$\therefore \text{IF} = e^{x^2}$$

$$\therefore z.e^{x^2} = \int x^3 e^{x^2} dx + c$$

$$\Rightarrow \tan y = ce^{-x^2} + \frac{e^{-x^2}}{2} \cdot e^{x^2} \{x^2 - 1\}$$

$$\Rightarrow \tan y = ce^{-x^2} + \frac{1}{2} \{x^2 - 1\}$$

$$\Rightarrow (AB)^2 = 1$$

4. Let $f: R \rightarrow R$ be a continuous function which satisfies $f(x) = \int_0^x f(t) dt$. Then the value of $f(\ln 5)$ is

[IIT-JEE 2009]

Sol. Answer (0)

$$f(x) = \int_0^x f(t) dt$$

$$\Rightarrow f'(x) = f(x)$$

$$\Rightarrow \frac{dy}{dx} = y \quad [y = f(x), \frac{dy}{dx} = f'(x)]$$

$$\Rightarrow \frac{dy}{dx} = \int 1 \cdot dx$$

$$\Rightarrow \ln y = x + \ln k$$

$$\Rightarrow \ln \frac{y}{k} = x$$

$$\Rightarrow y = ke^x$$

$$\Rightarrow f(x) = ke^x$$

$$\text{Now } f(x) = \int_0^x f(t) dt$$

$$\Rightarrow ke^x = \int_0^x ke^t dt = k \int_0^x e^t dt$$

$$\Rightarrow ke^x = k \left[e^t \right]_0^x$$

$$\Rightarrow ke^x = k[e^x - e^0]$$

$$\Rightarrow ke^x = ke^x - k$$

$$\Rightarrow k = 0$$

$$\therefore f(x) = 0 \quad \forall x$$

$$\Rightarrow f(\ln 5) = 0$$

5. Let f be a real-valued differentiable function on R (the set of all real numbers) such that $f(1) = 1$. If the y -intercept of the tangent at any point $P(x, y)$ on the curve $y = f(x)$ is equal to the cube of the abscissa of P , then the value of $f(-3)$ is equal to **[IIT-JEE 2010]**

Sol. Answer (9)

The equation of the tangent at (x, y) to the given curve $y = f(x)$ is

$$Y - y = \frac{dy}{dx}(X - x)$$

$$Y - \text{intercept} = y - x \frac{dy}{dx}$$

According to the question,

$$x^3 = y - x \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} - \frac{y}{x} = -x^2$$

which is linear in y

$$\text{I.F.} = e^{\int \frac{-1}{x} dx}$$

$$= \frac{1}{x}$$

Required solution is

$$y \frac{1}{x} = \int -x^2 \cdot \frac{1}{x} dx$$

$$\frac{y}{x} = \frac{-x^2}{2} + C$$

$$y = \frac{-x^3}{2} + Cx$$

at $x = 1, y = 1$

$$\therefore 1 = \frac{-1}{2} + C$$

$$\Rightarrow C = \frac{3}{2}$$

$$\text{Now, } f(-3) = \frac{27}{2} + \frac{3}{2}(-3)$$

$$= \frac{27 - 9}{2} = 9$$

6. Let $y'(x) + y(x)g'(x) = g(x)g'(x)$, $y(0) = 0$, $x \in \mathbb{R}$, where $f'(x)$ denotes $\frac{df(x)}{dx}$ and $g(x)$ is a given non-constant differentiable function on $\mathbb{R}$ with $g(0) = g(2) = 0$. Then the value of $y(2)$ is [IIT-JEE 2011]

Sol. Answer (0)

$$y'(x) + y(x)g'(x) = g(x) \cdot g'(x)$$

$\Rightarrow$ which is linear differential equation

$$\text{I.F.} = e^{\int g'(x) dx} = e^{g(x)}$$

Solution is

$$y(x)e^{g(x)} = \int e^{g(x)} \cdot g(x)g'(x) dx$$

$$y(x)e^{g(x)} = e^{g(x)}(g(x) - 1) + k$$

where k is a constant of integration

$$\text{For } x = 0, k = 1$$

$$\text{For } x = 2, y(2) = 0$$

7. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function with $f(0) = 0$. If $y = f(x)$ satisfies the differential equation $\frac{dy}{dx} = (2+5y)(5y-2)$, then the value of $\lim_{x \rightarrow -\infty} f(x)$ is _____. [JEE(Advanced)-2018]

Sol. Answer (0.40)

$$\frac{dy}{dx} = (2+5y)(5y-2)$$

$$\Rightarrow \int \frac{dy}{(5y+2)(5y-2)} = \int dx$$

$$\Rightarrow \frac{1}{4} \int \left(\frac{1}{5y-2} - \frac{1}{5y+2} \right) dy = \int dx$$

$$\Rightarrow \frac{1}{20} \ln \left| \frac{5y-2}{5y+2} \right| = x + C$$

$$f(0) = 0 \Rightarrow \ln \left| \frac{5y-2}{5y+2} \right| = x$$

$$x \rightarrow -\infty \Rightarrow \ln \left| \frac{5y-2}{5y+2} \right| \rightarrow -\infty$$

$$\Rightarrow \left| \frac{5y-2}{5y+2} \right| \rightarrow 0 \Rightarrow y \rightarrow \frac{2}{5}$$

$$\therefore \lim_{x \rightarrow -\infty} f(x) = \frac{2}{5} = 0.40$$

8. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function with $f(0) = 1$ and satisfying the equation $f(x+y) = f(x)f'(y) + f'(x)f(y)$ for all $x, y \in \mathbb{R}$. Then, the value of $\log_e(f(4))$ is _____. [JEE(Advanced)-2018]

Sol. Answer (2.00)

$$f(0) = 1, f: \mathbb{R} \rightarrow \mathbb{R}$$

$$f(x+y) = f(x)f'(y) + f'(x)f(y) \quad \forall x, y \in \mathbb{R}$$

$$\text{Put } x = y = 0 \Rightarrow f(0) = 2f(0)f'(0) \Rightarrow f'(0) = 1/2$$

$$\text{Put } y = 0,$$

$$f(x) = f(x)f'(0) + f'(x)f(0)$$

$$\Rightarrow f(x) = \frac{1}{2}f(x) + f'(x)$$

$$\Rightarrow \frac{f'(x)}{f(x)} = \frac{1}{2} \Rightarrow \ln(f(x)) = \frac{x}{2} + C$$

$$f(0) = 1 \Rightarrow C = 0$$

$$\Rightarrow \ln(f(x)) = \frac{x}{2}$$

$$\therefore \ln(f(4)) = \frac{4}{2} = 2$$

SECTION - G

Multiple True-False Type Questions

1. STATEMENT-1 : If m and n are respectively order and degree of differential equation, then m and n are mutually independent.

STATEMENT-2 : If general solution of a differential equation contains two arbitrary constants, then its order is 2.

STATEMENT-3 : The order and degree of differential equation $\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \left(x \frac{d^2y}{dx^2}\right)^{1/3}$ are 2 and 2 respectively.

(1) T F T

(2) T T T

(3) F F F

(4) F F T

Sol. Answer (2)

Statement-1 : True

Statement-2 : True

Statement-3 : True, Raising both sides to the power 6

$$\text{We have } \left(1 + \left(\frac{dy}{dx}\right)^2\right)^3 = x^2 \left(\frac{d^2y}{dx^2}\right)^2$$

It is clear that the order is 2 and the degree is also 2.

2. STATEMENT-1 : $y = e^x$ is a particular solution of $\frac{dy}{dx} = y$.

STATEMENT-2 : The differential equation representing family of curve $y = a \cos \omega t + b \sin \omega t$, where a and b are parameters, is $\frac{d^2y}{dt^2} - \omega^2 y = 0$.

STATEMENT-3 : $y = \frac{1}{2}x^3 + c_1x + c_2$ is a general solution of $\frac{d^2y}{dx^2} = 3x$.

(1) T F T

(2) T T T

(3) F F F

(4) F F T

Sol. Answer (1)

Statement-1 : True, $\frac{dy}{dx} = y \Rightarrow \int \frac{dy}{y} = \int 1 \cdot dx$

$$\ln y = x + \ln c$$

$$\Rightarrow y = ce^x$$

$$\text{For } c = 1, y = e^x$$

Statement-2 : False, $y = a \cos \omega t + b \sin \omega t$

$$\frac{dy}{dx} = -a\omega \sin \omega t + b\omega \cos \omega t$$

$$\frac{d^2y}{dx^2} = -a\omega^2 \cos \omega t - b\omega^2 \sin \omega t = -\omega^2 y$$

$$\therefore \frac{d^2y}{dx^2} + \omega^2 y = 0$$

Statement-3 : True

SECTION - H

Aakash Challengers Questions

1. If the differential equation satisfied by $y = A \sin(98x) + B \cos(98x)$ is $y_2 + cy = 0$ then the value of c is _____.

Sol. We have,

$$\Rightarrow y = A \sin(98x) + B \cos(98x)$$

$$\Rightarrow y_1 = 98A \cos(98x) - 98B \sin(98x)$$

$$\Rightarrow y_2 = -(98)^2 y \Rightarrow c = 98^2 = 9604$$

$$\Rightarrow c = 9604.$$

2. The differential equation of all straight lines which are at a fixed distance of 10 units from the origin is $(y - xy_1)^2 = A(1 + y_1^2)$ then A is equal to _____.

Sol. Let the equation of the line be $y = mx + c$

$$\Rightarrow \frac{dy}{dx} = m \quad \dots(i)$$

Also the distance of the line from $(0, 0)$ is 10

$$\Rightarrow \left| \frac{c}{\sqrt{1+m^2}} \right| = 10$$

$$c^2 = 100 (1 + m^2) \quad \dots(ii)$$

$$\Rightarrow c^2 = 100 \left(1 + \left(\frac{dy}{dx} \right)^2 \right)$$

But $c = y - mx = y - y_1x$, where $y_1 = \frac{dy}{dx}$

$$(y - xy_1)^2 = 100 (1 + y_1)^2$$

$$\Rightarrow A = 100$$

3. Let $\frac{dy}{dx} = \frac{y\phi'(x) - y^2}{\phi(x)}$, where $\phi(x)$ is a function satisfying $\phi(1) = 1$, $\phi(4) = 1296$. If $y(1) = 1$ then $y(4)$ is equal to _____

Sol. $\frac{dy}{dx} = \frac{y\phi'(x) - y^2}{\phi(x)}$

$$\Rightarrow \frac{dy}{dx} = \frac{y\phi'(x)}{\phi(x)} - \frac{y^2}{\phi(x)}$$

$$\Rightarrow \frac{1}{y^2} \frac{dy}{dx} = \frac{1}{y} \frac{\phi'(x)}{\phi(x)} - \frac{1}{\phi(x)}$$

$$\Rightarrow \frac{1}{y^2} \frac{dy}{dx} - \frac{1}{y} \frac{\phi'(x)}{\phi(x)} = -\frac{1}{\phi(x)}$$

Let $\frac{1}{y} = v$

$$-\frac{1}{y^2} \frac{dy}{dx} = \frac{dv}{dx} \Rightarrow -\frac{dv}{dx} - v \cdot \frac{\phi'(x)}{\phi(x)} = -\frac{1}{\phi(x)}$$

$$\Rightarrow \frac{dv}{dx} + v \frac{\phi'(x)}{\phi(x)} = \frac{1}{\phi(x)}, \text{ which is linear in } v$$

$$\text{I.F.} = e^{\int \frac{\phi'(x)}{\phi(x)} dx} = e^{\ln \phi(x)} = \phi(x)$$

The solution of the given differential equation is

$$v(\phi(x)) = \int \frac{1}{\phi(x)} \cdot \phi(x) dx$$

$$\Rightarrow v(\phi(x)) = x + c$$

$$\Rightarrow \frac{1}{y} (\phi(x)) = x + c$$

$$\text{at } x = 1, y = 1, \phi(1) = 1 \Rightarrow c = 0$$

$$\text{hence } \frac{1}{y} \phi(x) = x \Rightarrow y = \frac{\phi(x)}{x}$$

$$\text{Initial for } x = 4, \phi(4) = 1296$$

$$\Rightarrow y = \frac{1296}{4} = 324$$

4. If (2, 4) is a point on the orthogonal trajectory of $x^2 + y^2 - ay = 0$, then the orthogonal trajectory is a circle with radius _____

Sol. The differential equation of the given system of circles

$x^2 + y^2 - ay = 0$ is given by

$$2x + 2y \frac{dy}{dx} - \frac{x^2 + y^2}{y} \frac{dy}{dx} = 0$$

$$\Rightarrow 2x + \frac{y^2 - x^2}{y} \frac{dy}{dx} = 0 \quad \Rightarrow \quad \frac{dy}{dx} = \frac{2xy}{x^2 - y^2}$$

For orthogonal trajectory, we have

$$-\frac{dx}{dy} = \frac{2xy}{x^2 - y^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$$

Let us put $y = vx$ so that

$$v + x \frac{dv}{dx} = -\frac{v^2 - 1}{2v}$$

$$\Rightarrow x \frac{dv}{dx} = -\frac{v^2 - 1}{2v} \quad \Rightarrow \quad \ln x(1 + v^2) = \ln k$$

$$\Rightarrow x^2 + y^2 = kx$$

Which will pass through (2, 4) if $4 + 16 = 2k \Rightarrow k = 10$

The equation of orthogonal trajectory $x^2 + y^2 - 10x = 0$

This represents a circle of radius 5.

5. The value of $y(\sqrt{8}) + \frac{17}{9}$ if $(1 + x^2) \frac{dy}{dx} = x(1 - y)$, $y(0) = \frac{4}{3}$ is _____

Sol. We have, $(1 + x^2) \frac{dy}{dx} = x(1 - y)$

$$\Rightarrow \frac{dy}{1 - y} = \frac{x dx}{1 + x^2} \quad \Rightarrow \quad \frac{dy}{y - 1} = -\frac{x dx}{1 + x^2}$$

$$\Rightarrow \ln |y - 1| = -\frac{1}{2} \ln (1 + x^2) + c \quad \dots(i)$$

$$\text{When } x = 0, y = \frac{4}{3}$$

$$\Rightarrow \ln\left(\frac{4}{3}-1\right) = -\frac{1}{2}\ln(1+0) + c$$

$$c = \ln \frac{1}{3}$$

hence by equation (i)

$$\ln(y-1) = -\frac{1}{2}\ln(1+x^2) + \ln\left(\frac{1}{3}\right)$$

$$\Rightarrow y-1 = (1+x^2)^{-1/2} \left(\frac{1}{3}\right)$$

When $x = \sqrt{8}$

$$y-1 = (1+8)^{-1/2} \left(\frac{1}{3}\right)$$

$$\Rightarrow y-1 = \frac{1}{3} \times \frac{1}{3} = \frac{1}{9}$$

$$\Rightarrow y = \frac{1}{9} + 1 = \frac{10}{9}$$

$$\Rightarrow y + \frac{17}{9} = \frac{10}{9} + \frac{17}{9} = 3$$

6. The value of $y(\log 4)$ if $y_2 - 7y_1 + 12y = 0$, $y(0) = 2$, $y_1(0) = 7$ is _____

Sol. We have, $y_2 - 7y_1 + 12y = 0$

The given differential equation is $(D^2 - 7D + 12)y = 0$ is the solution of the given equation.

$D = 3, 4$

$$\text{Hence } y = Ae^{3x} + Be^{4x} \quad \dots(i)$$

Differentiating

$$y_1 = 3Ae^{3x} + 4Be^{4x} \quad \dots(ii)$$

at $x = 0$, $y = 2$

$$2 = A + B \quad \dots(iii)$$

at $x = 0$, $y_1 = 7$

$$7 = 3A + 4B \quad \dots(iv)$$

by (i) and (ii) $A = 1$, $B = 1$

$$\Rightarrow y = e^{3x} + e^{4x}$$

at $x = \log 4$

$$y = 4^3 + 4^4 = 64 + 256 = 320$$

7. An object falling from rest in air is subject not only to the gravitational force but also to air resistance. Assume that the air resistance is proportional to the velocity with constant of proportionality as $k > 0$, and acts in a direction opposite to motion ($g = 9.8 \text{ m/s}^2$). If the velocity can not exceed $A \text{ m/s}$ then the value of $10 Ak$ is

Sol. We have, $mg - kv = m \frac{dv}{dt}$

$$\Rightarrow g - \frac{k}{m}v = \frac{dv}{dt} \quad \Rightarrow \int \frac{dv}{g - \frac{k}{m}v} = \int dt$$

$$\Rightarrow \left(\frac{1}{-\frac{k}{m}} \right) \left[\ln \left(g - \frac{k}{m}v \right) \right] = [t]_0^t \Rightarrow \frac{-m}{k} \ln \left(\frac{g - \frac{k}{m}v}{g} \right) = t$$

$$\Rightarrow \ln \left(\frac{g - \frac{k}{m}v}{g} \right) = \frac{-kt}{m} \Rightarrow 1 - \frac{k}{m} \cdot \frac{v}{g} = e^{-\frac{kt}{m}}$$

$$\Rightarrow kv = mg (1 - e^{-kt/m})$$

$$\Rightarrow v_{\max} \text{ is at } e^{-kt/m} \rightarrow 0$$

$$v_{\max} = \frac{mg}{k}$$

$$\text{but } v_{\max} = A$$

$$\Rightarrow A = \frac{mg}{k} \Rightarrow Ak = mg$$

$$\text{But } m = 1 \Rightarrow Ak = g$$

$$\Rightarrow 10 Ak = 10g = 98$$

8. If the curve satisfying $(xy^4 + y)dx - xdy = 0$ passes through $(1,1)$ then the value of $-41(y(2))^3$ is _____

Sol. The given differential equation written as

$$xy^4 dx = xdy - ydx$$

$$\Rightarrow x^5 \cdot \frac{y^4}{x^4} \cdot \frac{1}{x^2} dx = \frac{xdy - ydx}{x^2} \Rightarrow x^3 dx = \frac{d\left(\frac{y}{x}\right)}{\left(\frac{y}{x}\right)^4}$$

$$\Rightarrow \frac{x^4}{4} = -\frac{x^3}{3y^3} + C$$

$$\text{Which will pass through } (1, 1) \text{ if } \frac{1}{4} + \frac{1}{3} = \frac{7}{12} = C$$

Thus the given curve is $\frac{x^4}{4} + \frac{x^3}{3y^3} = \frac{7}{12}$

For $x = 2$, we have

$$4 \frac{8}{3(y(2))^3} = \frac{7}{12}$$

$$\frac{8}{3(y(2))^3} = \frac{7}{12} - 4 = -\frac{41}{12}$$

$$\Rightarrow -41(y(2))^3 = 32$$

9. Solve the differential equation $\frac{dy}{dx} = y + \int_0^1 y(x) dx$, given that the value of y is 1, when $x = 0$.

Sol. $\frac{dy}{dx} = y + \int_0^1 y(x) dx \quad \dots(i)$

Differentiating both sides w.r.t. x

$$\frac{d^2y}{dx^2} = \frac{dy}{dx} + 0 \quad \left(\because \int_0^1 y(x) dx = \text{constant} \right)$$

$$\Rightarrow \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{dy}{dx} \Rightarrow \frac{dp}{dx} = p \quad \left(\because p = \frac{dy}{dx} \right)$$

$$\Rightarrow \frac{dp}{p} = dx$$

Integrating, we obtain

$\ln p = \ln k + x$, k being constant of integration

$$\frac{p}{k} = e^x \Rightarrow \frac{dy}{dx} = ke^x \Rightarrow dy = ke^x dx$$

Again integrating,

$$y = ke^x + k' : k' \text{ being another constant of integration}$$

Now, $y = 1$, when $x = 0$, so

$$1 = ke^0 + k' \Rightarrow 1 = k + k' \therefore k' = 1 - k$$

$$\text{Thus, } y = ke^x + 1 - k \quad \dots(ii)$$

Substituting (ii) in (i),

$$\frac{dy}{dx} = ke^x + 1 - k + \int_0^1 (ke^x + 1 - k) dx$$

$$\Rightarrow ke^x = ke^x + 1 - k + \left[ke^x + (1 - k)x \right]_0^1$$

$$\Rightarrow 0 = 1 - k + ke + (1 - k) - k$$

$$\Rightarrow 0 = 2 + ke - 3k \Rightarrow k(3 - e) = 2 \quad \therefore k = \frac{2}{3 - e}$$

Hence from (ii)

$$y = \frac{2}{3 - e} e^x + 1 - \frac{2}{3 - e} = \frac{2e^x}{3 - e} + \frac{1 - e}{3 - e}$$

$$\Rightarrow y = \frac{2e^x - e + 1}{3 - e} \text{ is the solution.}$$

10. Suppose $g(x)$ is a real valued differentiable function satisfying $g'(x) + 2g(x) > 1$. Then show that $e^{2x} \left(g(x) - \frac{1}{2} \right)$ is an increasing function.

Sol. $g'(x) + 2g(x) > 1$

Multiplying throughout by the IF of the corresponding differential equation $e^{\int 2dx} = e^{2x}$

We have,

$$e^{2x} g'(x) + 2g(x)e^{2x} > e^{2x}$$

$$\Rightarrow \frac{d}{dx}(e^{2x} g(x)) > e^{2x}$$

$$\Rightarrow \frac{d}{dx}(e^{2x} g(x)) > \frac{d}{dx} \left(\frac{1}{2} e^{2x} \right)$$

$$\Rightarrow \frac{d}{dx}(e^{2x} g(x)) - \frac{d}{dx} \left(\frac{1}{2} e^{2x} \right) > 0$$

$$\Rightarrow \frac{d}{dx} \left(e^{2x} g(x) - \frac{1}{2} e^{2x} \right) > 0 \Rightarrow \frac{d}{dx} \left(e^{2x} \left(g(x) - \frac{1}{2} \right) \right) > 0$$

$$\Rightarrow \frac{df}{dx} > 0, \text{ where } f = e^{2x} \left(g(x) - \frac{1}{2} \right)$$

Thus $f(x)$ is an increasing function, i.e., $e^{2x} \left(g(x) - \frac{1}{2} \right)$ is an increasing function.

11. Find the differential equation satisfied by family of parabolas $y^2 = 4a(x + a)$ and show that the orthogonal trajectory of the system belongs to system itself.

Sol. Differentiating $y^2 = 4a(x + a)$ w.r.t. x we have(i)

$$2y \frac{dy}{dx} = 4a$$

$$\therefore \frac{dy}{dx} = \frac{2a}{y} \quad \dots\dots(ii)$$

Eliminating a between (i) and (ii)

$$y^2 = 4 \left(\frac{y \frac{dy}{dx}}{2} \right) \left\{ x + \frac{y \frac{dy}{dx}}{2} \right\}$$

$$\Rightarrow y^2 = 2xy \frac{dy}{dx} + y^2 \left(\frac{dy}{dx} \right)^2$$

$$\Rightarrow y^2 \left(\frac{dy}{dx} \right)^2 + 2xy \frac{dy}{dx} - y^2 = 0 \quad \dots(A)$$

To get the orthogonal trajectory we replace $\frac{dy}{dx}$ by $-\frac{dx}{dy}$ in (A) to obtain.

$$y^2 \left(\frac{dx}{dy} \right)^2 - 2xy \frac{dx}{dy} - y^2 = 0$$

$$\Rightarrow y^2 - 2xy \left(\frac{dy}{dx} \right) - y^2 \left(\frac{dy}{dx} \right)^2 = 0$$

$$\Rightarrow y^2 \left(\frac{dy}{dx} \right)^2 + 2xy \frac{dy}{dx} - y^2 = 0 \quad \dots(B)$$

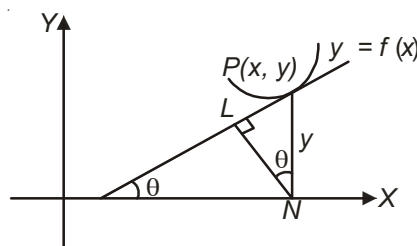
(B) is same as (A). Hence the solution of (A) and (B) are exactly the same families, which establish that the system is self orthogonal.

12. Find the curve in which the perpendicular from the foot of the ordinate to the tangent is of constant length.

Sol. NL = Length of perpendicular let fall from the foot of the ordinate to the tangent

$$= y \cos \theta = \frac{y}{\sec \theta}$$

$$= \frac{y}{\sqrt{1 + \tan^2 \theta}} = \frac{y}{\sqrt{1 + \left(\frac{dy}{dx} \right)^2}} \quad \left\{ \because \tan \theta = \frac{dy}{dx} \right\}$$



$$\text{Given } \frac{y}{\sqrt{1 + \left(\frac{dy}{dx} \right)^2}} = \text{constant} = k \text{ (say)}$$

$$\Rightarrow \frac{y^2}{k^2} = 1 + \left(\frac{dy}{dx} \right)^2 \Rightarrow \left(\frac{dy}{dx} \right)^2 = \frac{y^2 - k^2}{k^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{\sqrt{y^2 - k^2}}{k} \Rightarrow \frac{dy}{\sqrt{y^2 - k^2}} = \frac{1}{k} dx$$

Integrating we get, $\ln \left| y + \sqrt{y^2 - k^2} \right| = \frac{x}{k} + \ln c$; c being the constant of integration

$$\Rightarrow \frac{y + \sqrt{y^2 - k^2}}{e} = e^{\frac{x}{k}} \Rightarrow y + \sqrt{y^2 - k^2} = c \cdot e^{\frac{x}{k}}$$

$$\Rightarrow \left(y - c \cdot e^{\frac{x}{k}} \right) = -\sqrt{y^2 - k^2}$$

$$\Rightarrow \left(y - c \cdot e^{\frac{x}{k}} \right)^2 = y^2 - k^2$$

$$\Rightarrow y^2 + c^2 e^{\frac{2x}{k}} - 2c \cdot y e^{\frac{x}{k}} = y^2 - k^2$$

$$\Rightarrow c^2 e^{\frac{2x}{k}} - 2c y e^{\frac{x}{k}} + k^2 = 0 \text{ is the required family of curves.}$$

13. Solve $x \frac{dy}{dx} + y = y^2 \ln x$.

Sol. 1st Method

We have, $\frac{1}{y^2} \frac{dy}{dx} + \frac{1}{xy} = \frac{1}{x} \ln x$ (on dividing by xy^2)

Set $-\frac{1}{y} = v$, so that $\frac{1}{y^2} \frac{dy}{dx} = \frac{dv}{dx}$

Now the above equation becomes,

$$\frac{dv}{dx} - \frac{v}{x} = \frac{1}{x} \ln x, \text{ Which is linear in } v$$

$$\therefore \text{I.F.} = e^{\int -\frac{1}{x} dx} = e^{-\log x} = \frac{1}{x}$$

The solution is

$$V(\text{IF}) = \int \frac{1}{x} \ln x (\text{IF}) dx + k; k \text{ being the constant of integration}$$

$$\Rightarrow \frac{V}{x} = \int \frac{1}{x^2} \ln x dx + k \quad (\text{Setting } \ln x = t)$$

$$= -(t+1)e^{-t} + k$$

$$\Rightarrow \frac{V}{x} = -(1 + \ln x) \cdot \frac{1}{x} + k \quad (\because t = \ln x)$$

$$\Rightarrow -\frac{1}{xy} = -\frac{1}{x}(1+\ln x) + k \quad \left(\because V = \frac{1}{-y} \right)$$

$$\Rightarrow \frac{1}{xy} = \frac{1}{x}(1+\ln x) + C, C \text{ being another constant, is the general solution.}$$

2nd Method (by inspection)

$$x \frac{dy}{dx} + y = y^2 \ln x \Rightarrow xdy + ydx = (y^2 \ln x)dx$$

$$\Rightarrow d(xy) = y^2 \ln x dx$$

Dividing by $x^2 y^2$ we obtain

$$\frac{d(xy)}{x^2 y^2} = \frac{\ln x}{x^2} dx$$

$$\text{Integrating } \int \frac{d(xy)}{(xy)^2} = \int \frac{1}{x^2} \ln x dx = \int (\ln x) d\left(-\frac{1}{x}\right)$$

$$\Rightarrow -\frac{1}{xy} = \ln x \left(-\frac{1}{x}\right) - \int \frac{1}{x} \left(-\frac{1}{x}\right) dx$$

$$\Rightarrow \frac{1}{xy} = -\frac{\ln x}{x} - \frac{1}{x} + C \text{ being constant of integration}$$

$$\Rightarrow \frac{1}{xy} = \frac{\ln x}{x} + \frac{1}{x} + C \text{ being another constant is the general solution.}$$

14. Solve $\frac{dy}{dx} = \frac{1}{xy(x^2 \sin y^2 + 1)}$.

Sol. Given differential equation is not linear in $\frac{dy}{dx}$. So we try to find it. It is linear in $\frac{dx}{dy}$.

Inverting the given relation

$$\frac{dx}{dy} = xy(x^2 \sin y^2 + 1) = x^3 y \sin y^2 + xy$$

$$\frac{dx}{dy} - xy = (x^3 y \sin y^2)$$

Dividing by x^3 ,

$$\frac{1}{x^3} \frac{dx}{dy} - \frac{y}{x^2} = y \sin y^2 \quad \dots (A)$$

Let $-\frac{1}{x^2} = z \Rightarrow \frac{2}{x^2} \cdot \frac{dx}{dy} = \frac{dz}{dy}$

$\therefore$ (A) reduces to

$$\frac{dz}{dy} + 2yz = 2y \sin y^2, \text{ Which is linear}$$

$$\text{IF} = e^{\int 2y \, dy} = e^{y^2}$$

$\therefore$ Solution is given by

$$z(\text{IF}) = \int (2y \sin y^2)(\text{IF}) dy + k; k \text{ being constant of integration}$$

$$z(e^{y^2}) = \int 2y \sin y^2 \cdot e^{y^2} \cdot dy + k$$

$$= \int e^t \sin t \cdot dt + k \quad (\text{Setting } t = y^2)$$

$$ze^{y^2} = \frac{e^t}{2}(\sin t - \cos t) + k = \frac{e^{y^2}}{2}(\sin y^2 - \cos y^2) + k$$

$$\Rightarrow z = \frac{1}{2}(\sin y^2 - \cos y^2) + ke^{-y^2}$$

$$\Rightarrow -\frac{1}{x^2} = \frac{1}{2}(\sin y^2 - \cos y^2) + ke^{-y^2} \text{ is the general solution.}$$

15. Solve $\frac{dy}{dx} + 2 \cdot \frac{y}{x} = \frac{y^3}{x^3}$.

Sol. We offer three solutions to bring home the point that a differential equation can be solved by a variety of methods. Students should try to solve any equation by different methods so that they understand the strengths and weaknesses of each method.

1st method (by linear equation)

$$\frac{dy}{dx} + 2 \frac{y}{x} = \frac{y^3}{x^3} \Rightarrow \frac{1}{y^3} \frac{dy}{dx} + \frac{2}{x} \left(\frac{1}{y^2} \right) = \frac{1}{x^3} \quad \dots(i)$$

Set $-\frac{1}{y^2} = v$, so that $\frac{2}{y^3} \frac{dy}{dx} = \frac{dv}{dx}$

Equation (i) now reduces to

$$\frac{1}{2} \frac{dv}{dx} - \frac{2v}{x} = \frac{1}{x^3} \Rightarrow \frac{dv}{dx} - \frac{4v}{x} = \frac{2}{x^3}, \text{ which is linear}$$

$$\therefore \text{I.F.} = e^{\int -\frac{4}{x} dx} = e^{-4 \ln x} = \frac{1}{x^4}$$

Solution is

$$v(\text{I.F.}) = \int \frac{2}{x^3} (\text{I.F.}) dx + k ; k \text{ being constant of integration}$$

$$\Rightarrow v\left(\frac{1}{x^4}\right) = \int \frac{2}{x^3} \cdot \left(\frac{1}{x^4}\right) dx + k = \int \frac{2}{x^7} dx + k$$

$$\Rightarrow \frac{v}{x^4} = 2 \cdot \left(\frac{-1}{6x^6}\right) + k$$

$$\Rightarrow -\frac{1}{y^2 x^4} = -\frac{1}{3x^6} + k \Rightarrow \frac{1}{3x^6} - \frac{1}{y^2 x^4} = k$$

$$\Rightarrow (y^2 - 3x^2) = kx^6 y^2 \text{ is the general solution.}$$

2nd method (by homogeneous form)

As the equation is linear,

$$\frac{dy}{dx} = \frac{y^3}{x^3} - 2\frac{y}{x} = f\left(\frac{y}{x}\right), \text{ the equation is linear}$$

$$\text{We set } y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

Our equation now becomes

$$v + x \frac{dv}{dx} = v^3 - 2v$$

$$\Rightarrow x \frac{dv}{dx} = v^3 - 2v - v = v^3 - 3v = v(v^2 - 3)$$

$$\Rightarrow \frac{dv}{v(v^2 - 3)} = \frac{dx}{x} \Rightarrow \frac{1}{3} \left[\frac{v}{v^2 - 3} - \frac{1}{v} \right] dv = \frac{dx}{x} \quad (\text{By partial fractions})$$

Integrating, we obtain

$$\Rightarrow \frac{1}{3} \left[\frac{1}{2} \ln |v^2 - 3| - \ln |v| \right] + \ln k = \ln x$$

$$\Rightarrow \ln \frac{(v^2 - 3)^{\frac{1}{6}}}{v^{\frac{1}{3}}} \cdot k = \ln x$$

$$\Rightarrow x = \frac{(v^2 - 3)^{\frac{1}{6}}}{v^{\frac{1}{3}}} \cdot k \Rightarrow \frac{(y^2 - 3x^2)x^2}{x^2 y^2} \cdot k^6 = x^6$$

$$\Rightarrow x^6 y^2 = k^6 (y^2 - 3x^2)$$

$$\Rightarrow x^6 y^2 = C(y^2 - 3x^2)$$

C being another constant which is what we got earlier.

3rd solution (by inspection)

$$\frac{dy}{dx} + 2 \cdot \frac{y}{x} = \frac{y^3}{x^3} \Rightarrow xdy + 2ydx = \frac{y^3}{x^2} dx$$

$$\Rightarrow \frac{2ydx + xdy}{y^3} = \frac{dx}{x^2}$$

Dividing throughout by x^5

$$\frac{2ydx + xdy}{x^5 y^3} = \frac{dx}{x^7} \quad \dots(A)$$

Observe that $d\left(\frac{1}{x^4 y^2}\right) = -\left(\frac{4ydx + 2xdy}{x^5 \cdot y^3}\right)$

$$\Rightarrow \frac{2ydx + xdy}{x^5 y^3} = -\frac{1}{2} d\left(\frac{1}{x^4 y^2}\right)$$

Now from (A) using the above result

$$-\frac{1}{2} d\left(\frac{1}{x^4 y^2}\right) = \frac{dx}{x^7}$$

Integrating, we get,

$$-\frac{1}{2} \cdot \frac{1}{x^4 y^2} = \frac{-1}{6x^6} + k ; k \text{ being constant of integration}$$

$$\Rightarrow \frac{1}{6x^6} - \frac{1}{2x^4 y^2} = k$$

$$\Rightarrow \frac{y^2 - 3x^2}{6x^6 y^2} = k$$

$$\Rightarrow y^2 - 3x^2 = (6k)x^6 y^2$$

$$\Rightarrow y^2 - 3x^2 = k'x^6 y^2 \text{ is the general solution (} k' \text{ is another constant)}$$

16. Solve $\frac{dy}{dx} + y f'(x) = f(x) \cdot f'(x)$, where $f(x)$ is a given function.

Sol. $\frac{dy}{dx} + y f'(x) = f(x) \cdot f'(x)$, is of the form $\frac{dy}{dx} + P(x) \cdot y = Q(x)$ and thus a linear one.

$$\text{I.F.} = e^{\int f'(x) dx} = e^{f(x)}$$

The solution is

$$y \cdot (IF) = \int f(x) \cdot f'(x) (IF) \cdot dx + k; k = \text{being the constant of integration}$$

$$y \cdot e^{f(x)} = \int f(x) \cdot f'(x) \cdot e^{f(x)} \cdot dx + k$$

$$= \int t \cdot e^t dt + k \quad \{\text{set } t = f(x)\}$$

$$= e^t (t - 1) + k$$

$$= e^{f(x)} (f(x) - 1) + k$$

General solution is $y = f(x) - 1 + ke^{-f(x)}$

17. The family of curves, the subtangent at any point of which is the arithmetic mean of the co-ordinates of the point of tangency, is given by

$$(1) (x - y)^2 = cy$$

$$(2) (y - x)^2 = cx$$

$$(3) (x - y)^2 = cxy$$

$$(4) (x - y)^2 = cx^2 y^2$$

Sol. Let the family of the curves by $y = f(x)$

$$\tan \theta = \frac{I(PP')}{I(TP')} / [\text{subtangent}] = \frac{f(x)}{f'(x)}$$

$$\Rightarrow \frac{y}{y'} = \frac{x+y}{2} \quad (\text{given})$$

$$\Rightarrow y' = \frac{2y}{x+y}$$

$$\Rightarrow \frac{dy}{dx} = \frac{2y}{x+y} \Rightarrow \frac{dx}{dy} = \frac{x+y}{2y}$$

It is a homogeneous differential equation

Put $x = vy$

Differentiating w.r.t. y we get $\frac{dx}{dy} = v + y \frac{dv}{dy}$

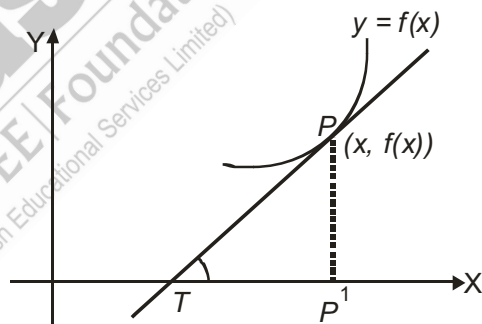
$$\Rightarrow v + y \frac{dv}{dy} = \frac{1+v}{2}$$

Simplifying we get

$$\frac{2}{1-v} dv = \frac{dy}{y}$$

$$\Rightarrow \log |y - x|^2 = \log |y| - \log c_1$$

$$\Rightarrow (x - y)^2 = cy$$



18. A line is drawn from a point $P(x, y)$ on the curve $y = f(x)$, making an angle with the x -axis which is supplementary to the one made by the tangent to the curve at $P(x, y)$. The line meets the x -axis at A . Another line perpendicular to it drawn from $P(x, y)$ meeting the y -axis at B . If $OA = OB$, where O is the origin, the equation of all curves which pass through $(1, 1)$ is

$$(1) \quad x^2 - y^2 + 2xy + 2 = 0$$

$$(2) \quad x^2 - y^2 + 2xy - 2 = 0$$

$$(3) \quad x^2 - y^2 + 2xy + 1 = 0$$

$$(4) \quad x^2 - y^2 + 2xy - 1 = 0$$

Sol. The equation of line through $P(x, y)$ making an angle with the x -axis which is supplementary to the angle made by the tangent at $P(x, y)$ is,

$$Y - y = \frac{dy}{dx}(X - x)$$

where it meets the x -axis.

$$Y = 0, X = x + \frac{y}{\frac{dy}{dx}} \Rightarrow OA = x + y \frac{dx}{dy}$$

The line through $P(x, y)$ and perpendicular to (i) is

$$Y - y = \frac{dx}{dy}(X - x)$$

where it meets the y -axis

$$X = 0, Y = y - x \frac{dx}{dy} \Rightarrow OB = y - x \frac{dx}{dy}$$

Since $OA = OB$,

$$\Rightarrow x + y \frac{dx}{dy} = y - x \frac{dx}{dy}$$

$$(y - x) = (y + x) \frac{dx}{dy}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y + x}{y - x}$$

Now it's a homogeneous differential equation.

Solving it and $(1, 1)$ satisfies so $x^2 + 2xy - y^2 = c$

19. The tangent and a normal to a curve at any point P meet the x and y axis at A, B, C and D respectively. Find the equation of curve passing through $(1, 0)$ if the centre of the circle through O, C, P and B lies on the line $y = x$ where O is the origin.

Sol. Let $P(x, y)$ be a point on the curve.

$$C \equiv \left(x + y \frac{dy}{dx}, 0 \right)$$

$$B \equiv \left(0, y - x \frac{dy}{dx} \right)$$

Circle passing through O, C, P and B has its centre at mid-point of BC .

Let the centre of the circle be (α, β)

$$2\alpha = x + y \frac{dy}{dx} \quad \text{and} \quad 2\beta = y - x \frac{dy}{dx}$$

And since $\beta = \alpha$, $y - x \frac{dy}{dx} = x + y \frac{dy}{dx}$

$$\frac{dy}{dx} = \frac{y-x}{y+x}$$

Let $y = vx$

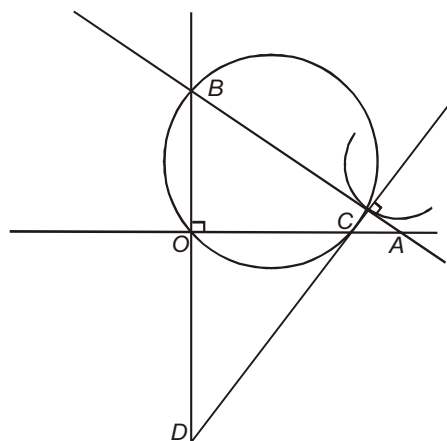
$$x \frac{dv}{dx} = -\frac{(1+v^2)}{1+v}$$

$$\int \frac{1+v}{v^2+1} dv = -\int \frac{dx}{x}$$

$$\log(\sqrt{x^2+y^2}) + \tan^{-1}\left(\frac{y}{x}\right) = c$$

As $x = 1$ and $y = 0$

So, the required curve is $\log(\sqrt{x^2+y^2}) + \tan^{-1}\left(\frac{y}{x}\right) = 0$



20. Given two curves $y = f(x)$ passing through $(0, 1)$ and $y = \int_{-\infty}^x f(t) dt$ passing through $(0, 1/n)$. The tangents drawn

to both the curves at the points with equal abscissas intersect on the x-axis, the curve is given by

(1) $y = e^{nx}$

(2) $y = nx$

(3) $y = n \ln x$

(4) $y = nx^2$

Sol. Equation of tangent to curve ; $y = f(x)$

$$(Y - y) = f'(x)(X - x)$$

Equation of tangent to the curve

$$g(x) = y_1 = \int_{-\infty}^x f(t) dt \text{ is}$$

$$(Y - y_1) = g'(x)(X - x) = f(x)(X - x)$$

Given that tangent with equal abscissae intersect on the x-axis

$$x - \frac{y}{f'(x)} = x - \frac{y_1}{f(x)}$$

$$\frac{f(x)}{f'(x)} = \frac{y_1}{f(x)}$$

$$\Rightarrow \frac{g'(x)}{g(x)} = \frac{f'(x)}{f(x)} \Rightarrow \frac{g'(x)}{g(x)} = k \Rightarrow g(x) = ce^{kx}$$

$$g'(x) = kce^{kx} \quad f(x) = kce^{kx}$$

$$y = f(x) \text{ passes through } (0, 1) \Rightarrow kc = 1 \Rightarrow f(x) = e^{nx}$$

